Unit 6 Bacteriology Part 3

Introduction

Upon completion of this course, the learner will have a basic understanding of the basic concepts of cultivation and identification of *Enterobacteriaceae,* other infectious gram-negative bacilli and Non-fermenting and Miscellaneous Gram-negative Bacilli

Agenda

This unit will take approximately 26 hours to complete

This unit contains the following lessons:

Lesson	Time
Bacteriology Part 3	10 Hours
Progress Check	1 Hour
Enterobacteriaceae Performance Test	5 Hour
Other gram-negative non-enteric bacilli Performance Test	5 Hours
Kirby-Bauer Performance Test	2 Hours
Bacteriology Case Study	1 Hour
Written Test 5 (Unit 6)	2 Hours

Lesson: **6.1 Enterobacteriaceae**

Lesson Objective: **6.1.1 Differentiate species of Enterobacteriaceae**

Topics:
General Characteristics
Opportunistic Members of the family *Enterobacteriaceae* and Associated Infections
Primary Intestinal Pathogens
Laboratory Diagnosis

Topic Objectives:
Discuss the general characteristics of the family *Enterobacteriaceae*
Describe opportunistic members of the family *Enterobacteriaceae* and associated infections
Discuss virulence factors of *Escherichia coli* strains
Describe primary intestinal pathogens of the family *Enterobacteriaceae* to include classification, species and sub species, virulence factors, antigenic structures, clinical infections, and biochemical testing
Discuss laboratory diagnosis of the family *Enterobacteriaceae* to include specimen processing, isolation and identification, pathogen screening, serologic grouping, and schema

Topic: General Characteristics

Topic Objectives: Discuss the general characteristics of the family *Enterobacteriaceae*

General Characteristics

- Enterics
  - All ferment glucose
  - All reduce nitrate to nitrites
  - All are oxidase negative
    - Except *Plesiomonas* spp.
  - Motility
    - All motile at body temperature except:
      - *Klebsiella* spp.
      - *Shigella* spp.
      - *Yersinia* spp.

- Microscopic and Colony Morphology
  - Gram-negative coccobacilli or straight rods
  - Facultatively anaerobic
  - Macroscopic morphology
    - Large, moist, gray colonies
    - Some are mucoid
      - *Klebsiella* spp.

- Growth on Media
  - MacConkey (MAC) agar
    - Is selective and differential media
    - Contains Bile salts and crystal violet that inhibit gram-positive
    - Lactose fermentation is differential
  - Eosin-methylene blue (EMB) agar
    - Selective and differential
    - Methylene blue inhibits gram-positive
    - Lactose and sucrose fermentation are differential
  - Hektoen enteric (HE) agar
    - Selective and differential
    - Bile salts inhibit gram-positive, some gram-negative
    - Lactose and sucrose fermentation are differential
      - Most non-pathogens ferment lactose and sucrose, showing an orange color (low pH)
      - Pathogens (*Salmonella* and *Shigella*) are green to blue green color with H_2S gas creating a black spot
  - Xylose lysine deoxycholate (XLD) agar
    - Selective (less so than MAC and HE) and differential
    - Yellow colonies
      - Fermenters or those not producing lysine decarboxylase *Escherichia coli, Citrobacter*
    - Colorless or red colonies
      - *Shigella*

        - Red colonies with black centers
            - Initially yellow then revert to red when lysine is decarboxylated, causing alkaline pH
                - *Salmonella*

- Virulence and Antigenic Factors
    - Adherence: prevent being removed from tissue
    - Toxins: food poisoning
    - Invasive enzymes: spreading factors
    - Antibiotic resistance
        - Extended spectrum β-Lactamases (ESBL)
    - Antigens
        - O antigen (somatic antigen)
            - Heat stable antigen located in the cell wall (LPS)
        - H antigen
            - Flagellar antigen; heat stable
        - K antigen
            - Capsular antigen; heat labile
            - K1 antigen of *E. coli*, Vi antigen of *Salmonella typhi*

- Biochemical Reactions
    - Certain key tests such as indole, methyl red, Voges-Proskauer, and citrate, known by the acronym IMViC, were routinely performed to group the most commonly isolated pathogens
    - Each IMViC reaction will be discussed for most isolates

- Clinical Significance
    - Two major types
        - Opportunistic pathogens
            - Normal flora
            - Infections in other “non-normal” sites
        - Primary pathogens
            - Generally, from ingesting contaminated food and water

Table 6.1 – Biochemical Characteristics of Tribes of Enterobacteriaceae

Tests or Substrate	Escherichieae	Edwardsielleae	Citrobacteriaceae	Salmonelleae*	Klebsiellae	Proteeae†	Yersiniae
H_2S (TSI agar)	–	+	+ or –	+	–	+ or –	–
Urease	–	–	(+^{w}) or –	–	– or (+)	+ or –	+
Indole	+ or –	+	– or +	–	–	+ or –	+ or –
Methyl red	+	+	+	+	–	+	+
Voges-Proskauer	–	–	–	–	+	–	–
Citrate (Simmons)	–	–	+	+	+	d	–
KCN	–	–	+ or –	–	+	+	–
Phenylalanine deaminase	–	–	–	–	–	+	–
Mucate	d	–		d	+ or –	–	
Mannitol	+ or –	–	+	+	+	– or +	+

Modified from Ewing WH: *Edwards and Ewing's identification of Enterobacteriaceae*, ed 4, East Norwalk, CT, 1986, Appleton & Lange.
H_2S, Hydrogen sulfide; *KCN*, potassium cyanide; *TSI*, triple sugar iron; +, ≥90% positive within 1 or 2 days; *(+)*, positive reaction after ≥3 days (decarboxylase tests: 3 or 4 days); –, ≥90% no reaction in 30 days; *+ or –*, most cultures positive, some strains negative; *– or +*, most strains negative, some cultures positive; *d*, different reactions, +, (+), –; +^{w}, weakly positive reaction.
**Salmonella* serovars Typhi and Paratyphi and some rare serovars fail to use citrate in Simmons medium. Cultures of serovar Paratyphi and some rare serotypes may fail to produce H_2S.
†Some cultures of *Proteus mirabilis* may yield positive Voges-Proskauer tests.

Table 6.2 – Bacterial Species and Infections They Commonly Produce

Bacterial Species	Diseases
Escherichia coli	Bacteriuria, septicemia, neonatal sepsis, meningitis, diarrheal syndrome
Shigella spp.	Diarrhea, dysentery
Edwardsiella spp.	Diarrhea, wound infection, septicemia, meningitis, enteric fever
Salmonella spp.	Septicemia, enteric fever, diarrhea
Citrobacter spp.	Opportunistic and hospital-acquired infections (wound, urinary)
Klebsiella spp.	Bacteriuria, pneumonia, septicemia
Enterobacter spp.	Opportunistic and hospital-acquired infection, wound infections, septicemia, bacteriuria
Serratia spp.	Opportunistic and hospital-acquired infection, wound infections, septicemia, bacteriuria
Proteus spp.	Bacteriuria, wound infection, septicemia
Providencia spp.	Opportunistic and hospital-acquired infection, wound infections, septicemia, bacteriuria
Morganella spp.	Opportunistic and hospital-acquired infections
Yersinia	
Y. pestis	Plague
Y. pseudotuberculosis	Mesenteric adenitis, diarrhea
Y. enterocolitica	Mesenteric adenitis, diarrhea
Erwinia spp.	Wounds contaminated with soil or vegetation
Pectobacterium spp.	Wounds contaminated with soil or vegetation

Modified from Washington J: *Laboratory procedures in clinical microbiology*, ed 2, New York, 1981, Springer-Verlag.

Topic: Opportunistic Members of the family Enterobacteriaceae and Associated infections

Topic Objective: Describe opportunistic members of the family Enterobacteriaceae and associated infections

Escherichia coli

- General Characteristics
  - Dry pink colony with a halo on MAC plates
  - Green metallic sheen on eosin methylene blue (EMB)
  - β-hemolysis on blood agar plate (BAP)
  - Motile
  - Sex pili, fimbriae
  - Possess O, H, K antigens

- Biochemical reactions of *E. coli*
  - IMVIC reaction + + - -
    - Indole: positive
    - Methyl red (MR): positive
    - Voges-Proskauer (VP): negative
    - Citrate: negative
  - Fermentation of glucose, lactose, trehalose, and xylose
  - Does not produce H_2S, DNase, urease, or PAD

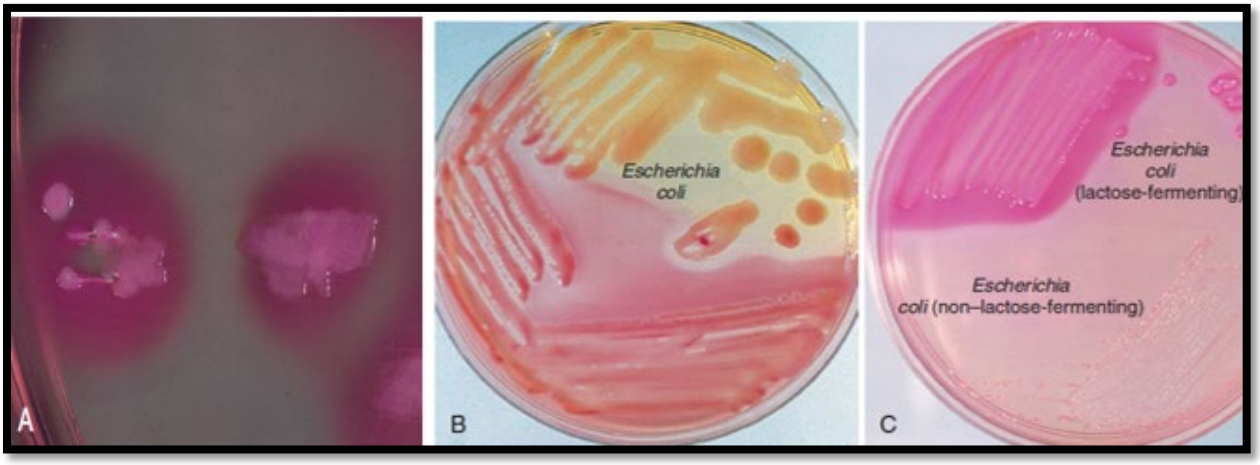


Figure 6.1 – Colonies of E. coli. A) Typical dry, lactose-positive Escherichia coli growing on MacConkey (MAC) agar. Note the pink precipitate surrounding the individual colonies. B) Mucoid colonies of E. coli growing on MAC agar. C) Non–lactose-fermenting E. coli compared with typical lactose-fermenting E. coli on MAC agar.

- **Clinical Types of *E. coli***
    - Uropathogenic *E. coli*
        - Most common cause of UTIs in humans
            - Pili allows attachment to epithelium
            - Cytolysins can kill immune cells and inhibit phagocytosis
            - Aerobactin allows bacteria to chelate iron.
    - Gastrointestinal pathogens
        - Five major categories
            - Enterotoxigenic (ETEC)
            - Enteropathogenic (EPEC)
            - Enteroinvasive (EIEC)
            - Enterohemorrhagic (EHEC)
            - Enteroadherent
                - Diffusely adherent (DAEC)
                - Enteroaggregative (EAEC)

- ETEC
    - Causes diarrhea in infants and adults especially in tropical and subtropical climates
    - High infective dose (10^6 to 10^{10} organisms) is required to initiate disease in an immunocompetent host
    - Infection is spread commonly via consumption of contaminated food or water
    - Last 1 to 5 days
    - Toxins released:
        - Heat labile toxin (LT): A and B subunits; B binds GM1 ganglioside receptor, and A acts on adenyl cyclase, converting adenosine triphosphate (ATP) to cyclic adenosine monophosphate (cAMP), which leads to hypersecretion of fluids
            - Heat-stable toxin (ST) stimulates guanylate cyclase, increasing cyclic guanosine monophosphate (cGMP), leading to hypersecretion
    - Protective mechanisms such as stomach acidity have been described as inhibiting colonization and initiation of disease
    - Usually a mild, self-limiting disease
    - Symptoms:
        - Watery diarrhea, abdominal cramps, and sometimes nausea, usually with no vomiting or fever

- EPEC
    - Cause infantile diarrhea in children less than 1 year
    - Outbreaks have occurred in nurseries and daycares
    - Symptoms:
        - Low-grade fever, malaise, vomiting, diarrhea
            - Large amounts of mucus without apparent blood
    - Serologic typing to identify EPEC

- EIEC
  - Affects both adults and children alike
    - U.S. and Worldwide infections rare
    - Direct transmission of EIEC from person to person via the fecal-oral route has been reported
  - Strains produce dysentery with direct penetration, invasion, and destruction of intestinal mucosa
  - Symptoms:
    - Fever, severe abdominal cramps, malaise, and watery diarrhea

- EHEC (*E. coli* 0157:H7)
  - The EHEC strain serotype O157:H7 causes hemorrhagic diarrhea, colitis
    - Watery diarrhea progressing to bloody diarrhea without pus can be fatal in young children and elderly
  - Also causes hemolytic-uremic syndrome (HUS)
  - Transmitted by undercooked meats, unpasteurized milk, apple cider, bean sprouts, and spinach
  - Symptoms:
    - Bloody diarrhea with abdominal cramps and low-grade fever or an absence of fever.
    - The stool contains no leukocytes, which distinguishes it from dysentery caused by *Shigella* spp. Or EIEC infections
  - Cytotoxins produced by *E. coli* O157:H7
    - Verotoxin I (phage mediated) cytotoxin
      - Damages Vero cells (African green monkey kidney cells)
      - Identical to the Shiga toxin (Stx) produced by Shigella dysenteriae type I
      - Toxin neutralized by Shiga toxin antibodies
    - Verotoxin II
      - Not neutralized by Shiga toxin antibodies
      - Biologically similar to, but immunologically different from Stx and Verotoxin I

- Laboratory Diagnosis *E. coli* O157:H7
  - Primary screening on a stool culture
    - MAC agar with sorbitol (SMAC) plate
      - *E. coli* O157:H7 does not ferment sorbitol in 48 hours, a characteristic that differentiates it from most other *E. coli*
      - *E. coli* O157:H7 appears colorless on SMAC agar, whereas most other strains produce pink colonies for sorbitol fermentation
  - Assay Testing
    - 4-methylumbelliferyl-β-D-glucuronide (MUG) assay
    - Enzyme-linked immunosorbent assay (ELISA) or latex agglutination
    - Culture assays and anti- body titers are also used to detect Vero cells in stool filtrates

- Enteroadherent *E. coli*
  - DAEC (Diffusely adherent *E. coli*)
    - Associated with UTI's and diarrheal disease
    - Uropathogenic DAEC strains are closely associated with cystitis in children and acute pyelonephritis in pregnant women

- EAEC (Enteroaggregative *E. coli*)
  - Occurs mostly in infants, children, and HIV patients
  - Causes diarrhea by adhering to the surface of the intestinal mucosa
  - Symptoms:
    - Watery diarrhea, vomiting, dehydration, and occasionally abdominal pain
    - Symptoms typically persist for at least 2 or more weeks

- Extraintestinal Infections of *E. coli*
  - Septicemia and meningitis
    - Most common in neonates and very young children
    - Gain infection just before or during delivery, and when infections involve the amniotic fluid
    - Associated with the capsule antigen K1
    - *E. coli* remains as a clinically significant isolate in blood cultures from adults

Other *Escherichia* Species

- *E. hermannii*
  - Yellow-pigmented colonies
    - Isolated from cerebrospinal fluid (CSF), wounds, and blood
    - Also found in the same sources as *E. coli* O157:H7
    - Clinical significance is still not fully established

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- *E. vulneris*
  - Also produces yellow colonies
  - Isolated from humans with infected wounds

- *E. albertii*
  - The newest species added to this genus
  - Associated with diarrheal disease in children

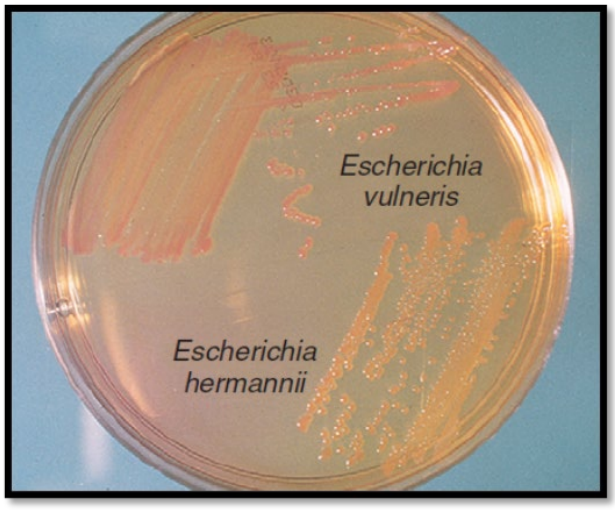


Figure 6.2 – Shows the yellow pigment of E. hermannii; E. vulneris can also produce a yellow pigment; but yellow is more prevalent to E. hermannii

Klebsiella, Enterobacter, Serratia, Pantoea, Cronobacter,* and *Hafnia

- General characteristics
  - Members of these genera belong to the Tribe *Klebsielleae*
  - Members of these genera are usually found in the intestinal tract of humans and animals or free-living in soil, water, and on plants
  - Associated with opportunistic and nosocomial infections

- Virulence factors of *Klebsiella* spp.
  - Polysaccharide capsule (occasionally evident in direct smears)
    - Prevents phagocytosis and some antimicrobials

- Common biochemical characteristics
  - Potassium cyanide broth negative
  - No H_2S production
  - Motility variable
  - Generally, urease negative
    - Few hydrolyze urea slowly
  - IMVIC: - - ++ (few exceptions on indole)
    - Except *K. oxytoca*: + - + +

Klebsiella pneumoniae

- Most common isolate

- Produces moist, gray, mucoid colonies on BAP

- Produces moist, mucoid, pink colonies on MAC

- Frequent cause of lower respiratory tract infections among hospitalized patients and in immuno-compromised hosts
  - Also found in wounds, UTIs, liver abscesses, and bacteremia

- Antimicrobial resistance
  - Severe due to the presence of the *K. pneumoniae* carbapenemase

Klebsiella pneumoniae Subspecies

- *K. pneumoniae* subspecies *ozaenae*
  - Isolated from nasal secretions and cerebral abscesses
  - Causes atrophic rhinitis, a tissue-destructive disease restricted to the nose
  - Antimicrobial resistance
    - Presence of plasmid-mediated ESBLs as led to large numbers of antimicrobial-resistant hospital-acquired infections

- *K. pneumoniae* subspecies *rhinoscleromatis*
  - Organism primarily found in Africa and South America

Table 6.3 – Differentiation of Common Species within the Genus Klebsiella

	K. pneumoniae subsp. *pneumoniae*			*K. oxytoca*			*K. pneumoniae* subsp. *ozaenae*		
Test or Substrate	**Sign**	**% +**	**(% +)**	**Sign**	**% +**	**(% +)**	**Sign**	**% +**	**(% +)**
Urease	+	95.4	(0.1)	+	90		d	0	(14.8)
Indole	–	0		+	99		–	0	
Methyl red	– or +	10		–	20		+	97.7	
Voges-Proskauer	+	98		+	96		–	0	
Citrate (Simmons)	+	98	(0.6)	+	95		d	30	(32.4)
Gelatin (22°C)	–	0	(0.2)	–	0		–	0	
Lysine decarboxylase	+	98	(0.1)	+	99		– or +	40	(6.3)
Malonate	+	92.5		+	98		–	6	
Mucate	+	90		+	93		– or +	25	
Sodium alginate (utilization)	+ or (+)	88.5	(9.2)	nd			– or (+)	0	(11)
Gas from glucose	+	96		+	97		d	50	(9.4)
Lactose	+	98.7	(1)	+	100		d	30	(61.3)
Dulcitol	– or +	30		+or–	55		–	0	
Organic acid media									
Citrate	+ or –	64.4		nd			– or +	18	
D-Tartrate	+ or –	67.1		nd			– or +	39	

Modified from Ewing WH: *Edwards and Ewing's identification of Enterobacteriaceae*, ed 4, East Norwalk, CT, 1986, Appleton and Lange.
+, ≥90% positive within 1 or 2 days; *(+)*, positive reaction after ≥3 days (decarboxylase tests: 3 or 4 days); –, ≥90% no reaction in 30 days; + *or* –, most cultures positive, some strains negative; – *or* +, most strains negative, some cultures positive; *d*, different reactions, +, (+), –; *nd*, no data.

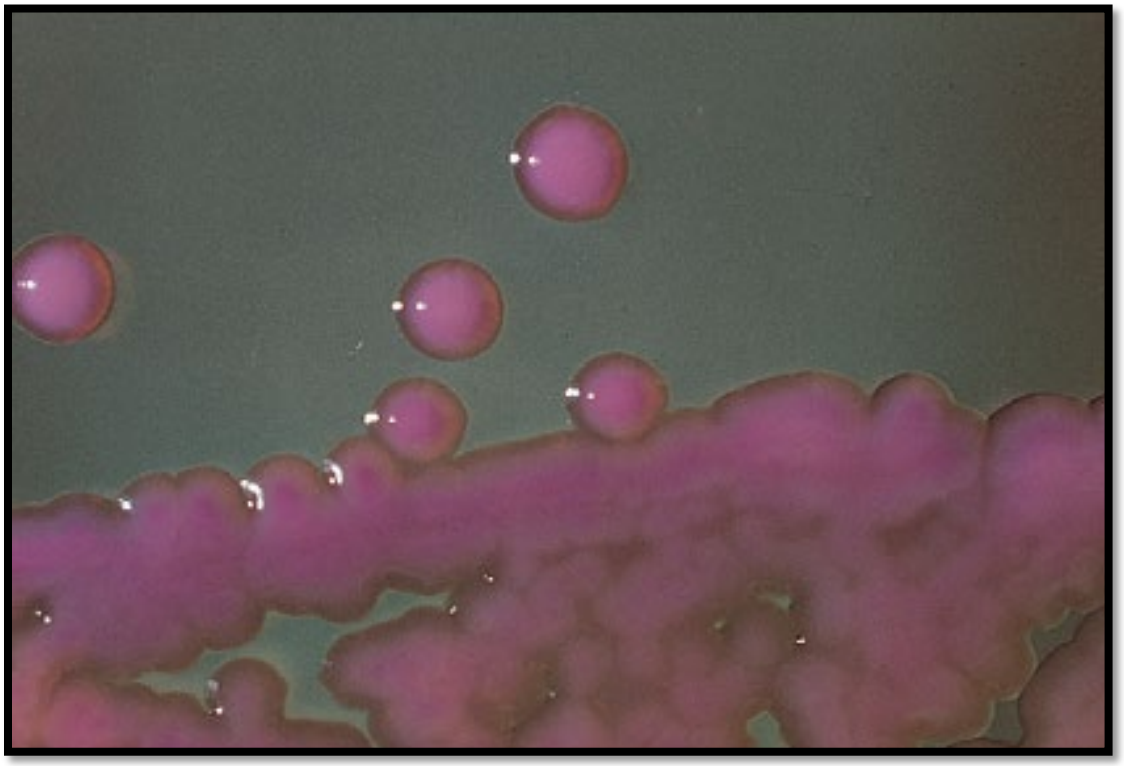


Figure 6.3 – The mucoid appearance of K. pneumoniae on MAC agar

K. oxytoca

- Similar to *K. pneumoniae* except indole positive
- Affects similar sites
- Isolates have been linked to antibiotic-associated hemorrhagic colitis

***Raoutella* Species**

- Two species have been isolated in urine, respiratory tracts, and blood
  - *R. (Klebsiella) ornithinolytica*
    - Indole and ornithine decarboxylase positive
  - *R. (Klebsiella) planticola*

***Enterobacter* species**

- Isolated from wounds, urine, blood, and CSF
- Two clinically significant species most isolated in clinical specimens
  - *E. cloacae*
  - *E. aerogenes*
- *E. gergoviae*
  - Respiratory samples, rarely blood
- *E. hormaechei*
- Colony morphology and Biochemical reactions of *Enterobacter* spp.
  - Resembles *Klebsiella* when growing on MAC agar
  - IMVIC reactions: - - + +

***Pantoea* and *Cronobacter* Species**

- *Pantoea agglomerans*
  - Formerly *Enterobacter agglomerans*
  - Similar to *Klebsiella pneumoniae*, so it needs to differentiate
  - Primarily a plant pathogen

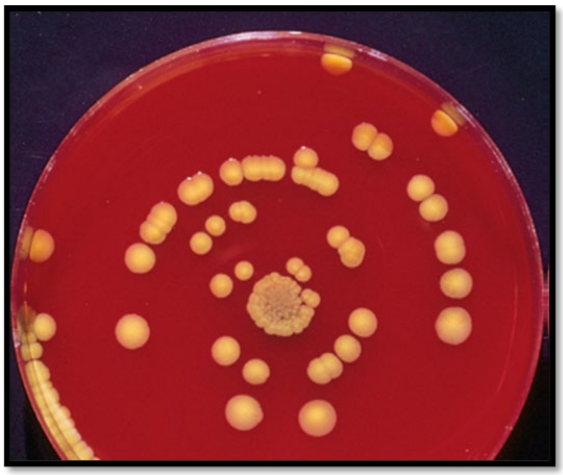


Figure 6.4 – Yellow-pigmented P. agglomerans on SBA

- *Cronobacter sakazakii*
    - Colony morphology
        - Colonies produce a yellow pigment
    - Isolated in case of meningitis and bacteremia in neonates
        - Due to contaminated powdered infant formula

      Also isolated from brain abscesses, respiratory and wound infections

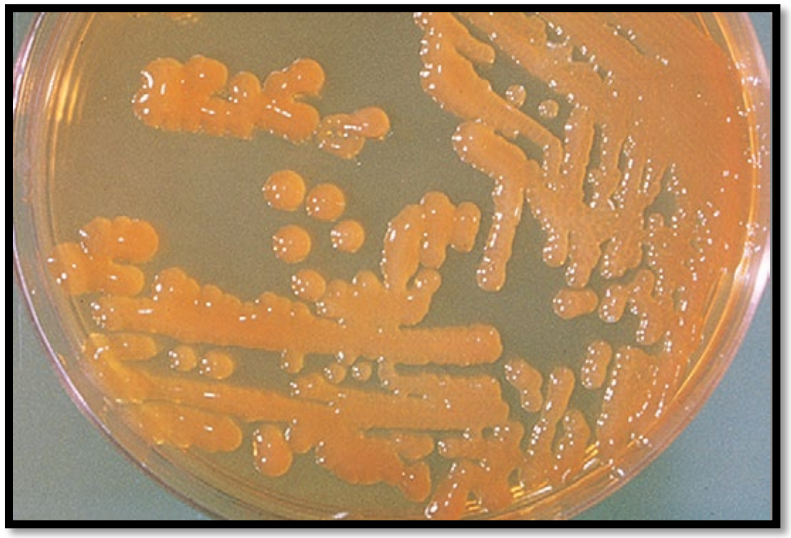


Figure 6.5 – Mucoid, yellow-pigmented colonies of C. sakazakii growing BHI agar

Serratia Species

- General characteristics of *Serratia* spp.
    - Positive ONPG (slow lactose)
        - Exception is *S. fonticola*
    - Produces extracellular DNase
    - Opportunistic pathogens associated with outbreaks in health care settings

- Colony morphology and Biochemical reactions of *Serratia* spp.
    - Colony morphology
        - Colonies produce a pink-to-red pigment (prodigiosin) when incubated at room temperature
            - Pigment production is typically a characteristic in those strains of environmental origin
    - IMVIC reactions: - - + +

- Antimicrobial Susceptibility of *Serratia* spp.
    - Highly resistant to antimicrobials
        - Susceptibility tests must be performed on each isolate to determine appropriate antimicrobial therapy

Serratia marcescens

- Considered the most clinically significant of the species
- Found frequently in hospital acquired infections of the urinary or respiratory tract and in bacteremic outbreaks in nurseries and cardiac surgery and burn units

Serratia rubidaea* and *Serratia plymuthica

- Also produce a pink-to-red pigment when incubated at room temperature

Serratia odorifera

- Emits a dirty, musty odor resembling that of rotten potatoes
- Contains two bio groups
    - Bio group 1
        - Isolated predominantly from the respiratory tract
        - Positive for sucrose, raffinose, ornithine
            - Most positive for indole (60%)
    - Bio group 2
        - Isolated from blood and CSF
        - Negative for sucrose, raffinose, ornithine
            - Most positive for indole (50%)

Other *Serratia* spp.

- *S. ficaria*
- *S. entomophila*
- *S. fonticola*



Figure 6.6 – (L)-S. marcescens growing on CHOC agar. (R)- Showing brick-red pigment when grown on MAC agar

Hafnia

- The genus *Hafnia* is composed of one species *Hafnia alvei*
    - Isolated from many anatomic sites in humans and in the environment
    - Linked to gastroenteritis and is occasionally isolated from stool cultures
    - Two distinct biotypes are recognized:
        - *H. alvei*
        - *H. alvei* biotype 1
            - Grows in the beer wort of breweries
            - Has not been isolated clinically

***Proteus* Species**

- General characteristics
    - The genera *Proteus, Morganella*, and *Providencia* belong to the tribe Proteeae
    - They are widely disseminated in the environment, are normal intestinal microbiota, and are recognized as opportunistic pathogens isolated from urine, wounds, and ear, and bacteremia infections
        - Thought to play a role in the ascending nature of *Proteus*-associated UTIs

- *P. mirabilis* and *P. vulgaris*
    - *P. mirabilis* most commonly isolated
        - Both species, particularly *P. mirabilis*, can produce “swarming” colonies on nonselective media, such as SBA

- Colony morphology of *Proteus* spp.
    - Does not ferment lactose
    - Produce “swarming” colonies on nonselective media, such as SBA
        - Due to hyperflagellated, elongated, polyploid cells
    - Has a distinct odor described as “burnt chocolate”

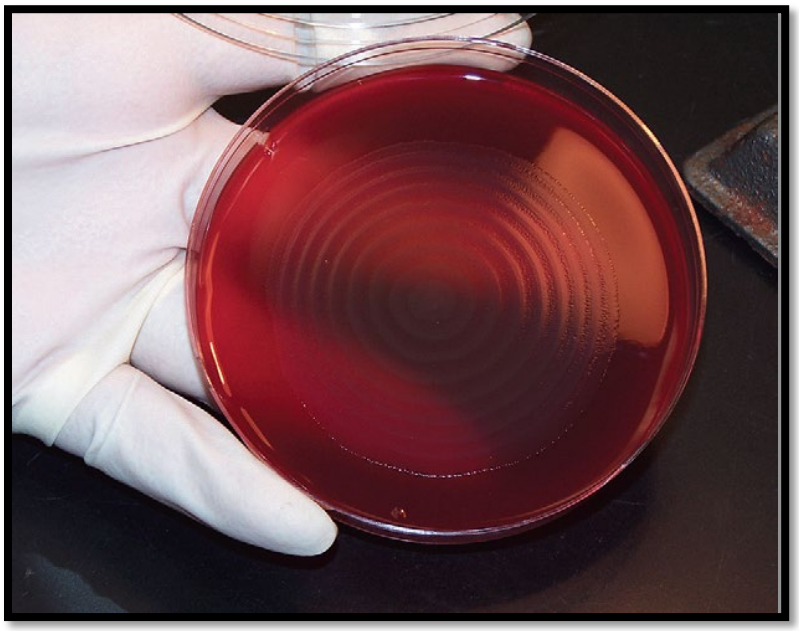


Figure 6.7 – Proteus mirabilis swarming on sheep blood agar

- Biochemical Reactions of *Proteus* spp.
    - Distinguished from the other members of the Enterobacteriaceae by the ability to deaminate the amino acid phenylalanine
    - IMVIC reactions
        - *P. mirabilis*: + + - -
        - *P. vulgaris*: - + - -
    - Produce H_2S
    - Urease positive

***Morganella* Species**

- The genus *Morganella* has only one species:
    - *M. morganii*
        - Isolated in UTI infections and identified as a cause of neonatal sepsis

***Providencia* Species**

- The genus Providencia consists of five species:
    - *P. alcalifaciens*
    - *P. stuartii*
    - *P. rettgeri*
    - *P. rustigianii*
    - *P. heimbachae*

- Two major pathogens:
    - *P. Rettgeri*
        - Known urinary tract pathogen
        - Occasionally isolated in nosocomial outbreaks
    - *P. Stuartii*
        - Implicated in nosocomial outbreaks in burn unit outbreaks
        - Also isolated in urine cultures

- Possibility of a third pathogen in the genus:
    - *P. alcalifaciens*
    - Role as a possible cause of diarrhea in children has not been proven

- Biochemical Reactions
    - IMVIC reactions: + + - +

- Difficult to treat due to high resistance to antimicrobials

Table 6.4 – Differentiating Characteristics of Selected Species of Proteus, Providencia, and Morganella

Test	*Proteus penneri*	*Proteus mirabilis*	*Proteus vulgaris*	*Providencia alcalifaciens*	*Providencia stuartii*	*Providencia rettgeri*	*Morganella morganii*
Indole	–		+	+	+	+	+
Methyl red	+	+	+	+	+	+	+
Voges-Proskauer	–	– or +	–	–	–	–	–
Simmons citrate	–	+ or (+)	d	+	+	+	–
Christensen urea	+	+ or (+)	+	–	– or +	+	+
H_2S (TSI)	– (70%)	+	+	–	–	–	–
Ornithine decarboxylase	–	+	–	–	–	–	+
Phenylalanine deaminase	+	+	+	+	+	+	+
Acid produced from							
Sucrose	+	d	+	d	d	d	–
Mannitol	–	–	–	–	d	+	–
Salicin	–	–	d	–	–	d	–
Adonitol	–	–	–	+	–	+	–
Rhamnose	–	–	–	–	–	+ or –	–
Maltose	+	–	+	–	–	–	–
Xylose	+	+	+ or (+)	–	–	– or +	–
Arabitol	–	–	–	–	–	+	–
Swarms	+	+	+	–	–	–	–

Modified from Washington J: *Laboratory procedures in clinical microbiology*, ed 2, New York, 1981, Springer-Verlag.
H_2S, Hydrogen sulfide; *TSI*, triple-sugar iron agar; *+*, ≥90% positive reaction within 1 or 2 days; *–*, no reaction (≥90%) in 30 days; *– or +*, most strains negative, some cultures positive; *+ or (+)*, most reactions occur within 1 or 2 days, some are delayed; *d*, different reactions, + or –, most cultures positive, some strains negative.

Edwardsiella

- The genus *Edwardsiella* is composed of three species:
  - *E. tarda*
  - *E. hoshinae*
  - *E. ictaluri*

- Isolated in bacteremia and wound infections

- *E. tarda* is the only pathogen
  - Opportunistic pathogen

- Biochemical identification of *E. tarda*
  - IMVIC: + + - -
  - Urea negative
  - Lysine decarboxylase positive
  - H_2S positive
  - Indole positive
  - Does not grow on Simmon's citrate

Citrobacter Species

- General characteristics
  - Considered inhabitants of the GI tract and are associated with hospital-acquired infections, most frequently UTIs
  - Consists of 11 species that have been isolated from clinical specimens
    - The three species most often isolated are:
      - *C. freundii*
      - *C. koseri*
      - *C. braakii*

- Biochemical Reactions of *Citrobacter* spp.
    - Biochemically similar to *Salmonella*
    - IMVIC reactions: + + - +
    - All species citrate positive
        - Hence the name
    - Weak urease activity
    - Ferment lactose
        - Resemble *E. coli*
    - Most produce H_2S

Citrobacter freundii

- *Associated with infectious diseases acquired in hospital settings; UTIs, pneumonias, and intraabdominal abscesses have been reported*

- *Also has been associated with endocarditis in intravenous drug abuser*

- *C. freundii on primary selective media can be mistaken for Salmonella*
    - *Some C. freundii produce H_2S and some strains (50%) fail to ferment lactose*
    - *Very important to differentiate in stool cultures*

Citrobacter koseri

- Documented as the cause of nursery outbreaks of neonatal meningitis and brain abscesses

Topic: Primary Intestinal Pathogens of the Family Enterobacteriaceae

Topic Objective: Describe primary intestinal pathogens of the family Enterobacteriaceae to include classification, species and sub species, virulence factors, antigenic structures, clinical infections, and biochemical testing

***Salmonella* species**

- General characteristics
  - Only two species, *S. enterica* (the type species of the genus) and *S. bongori* (rare), are now designated in the genus
  - Facultative anaerobic

- Colony Morphology of *Salmonella* spp.
  - Colony morphology
    - Clear, colorless, non–lactose-fermenting colonies with black centers on MAC agar
    - Colonies will have black centers on media containing indicators for H_2S production (e.g., HE or XLD)

- Biochemical Reactions of *Salmonella* spp.
  - IMVIC reactions: - + - +
  - Negative for phenylalanine deaminase
  - Negative for urease
  - Most produce H_2S
  - Voges-Proskauer (VP) negative
  - Indole negative
    - Exception: *Salmonella paratyphi* A
  - No growth in media containing potassium cyanide

Salmonella enterica

- Six subspecies in *S. enterica*
  - *S. enterica subsp. enterica (also called subspecies I)*
  - *S. enterica subsp. salamae (subspecies II),*
  - *S. enterica subsp. arizonae (subspecies IIIa)*
  - *S. enterica subsp. diarizonae (subspecies IIIb)*
  - *S. enterica subsp. houtenae (subspecies IV)*
  - *S. enterica subsp. Indica*

- Nearly all former *Salmonella* spp. Have been placed as serotypes below the level of *S. enterica subsp. enterica* Typhi
  - Simply written as *Salmonella* Typhi (serotype is capitalized and not italicized)

- Differentiating *Salmonella* subgroups and members
  - Lysine decarboxylase rules out *S. paratyphi*
  - Ornithine decarboxylase rules out *S. typhi*
  - Trehalose fermentation rules out *S. choleraesuis*

Table 6.5 – Biochemical Differentiation of Selected Members of the Genus Salmonella

Test	*S.* serotype Choleraesuis	*S.* serotype Paratyphi	*S.* serotype Typhi	Other*
Arabinose fermentation	–	+	–	+
Citrate utilization	V	–	–	+
Glucose gas production	+	+	–	+
H_2S (TSI)	V	–	+	+
Lysine decarboxylase	+	–	+	+
Ornithine decarboxylase	+	+	–	+
Rhamnose fermentation	+	+	–	+
Trehalose fermentation	–	+	+	+

Data from Farmer JJ, et al: Enterobacteriaceae: introduction and identification. In Murray PR, et al, editors: *Manual of clinical microbiology*, ed 9, Washington, DC, 2007, ASM Press.
H_2S, Hydrogen sulfide; *TSI,* triple sugar iron agar; –, ≤9% of strains positive; +, ≥90% of strains positive; *V,* 10% to 89% of strains positive.
*Typical strains in serogroups A through E.

- Virulence factors of *S. enterica*
  - Fimbriae required for adhesion to intestine
  - Have ability to traverse intestinal mucosa
  - Enterotoxin
    - Involved in gastroenteritis
  - Note: Factors responsible for the virulence of salmonellae have been the subject of speculation and are still being investigated

- Antigenic Structures of *S. enterica*
  - H antigen
    - Antigens occur in phases
    - Phase I - the specific phase
    - Phase II – the non-specific phase
  - O antigen - Heat stable antigen is the lipopolysaccharide located in the outer membrane of the cell wall
  - K capsule antigen
    - Designated as the Vi antigen. The serologic identification of the Vi antigen is important in identifying *Salmonella* serotype typhi
    - Prevents phagocytosis

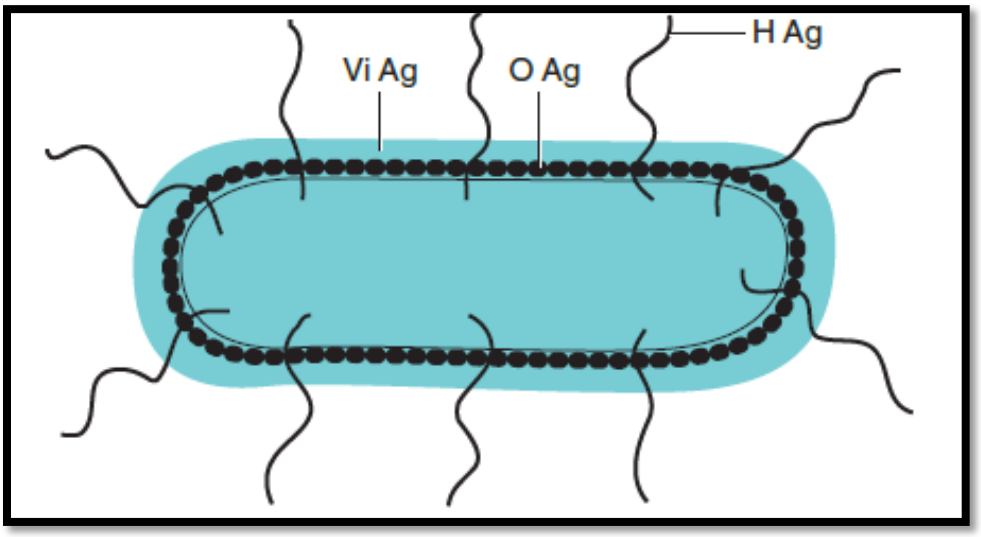


Figure 6.8 – Antigenic Structures of Salmonellae Used in Serologic Typing

- Clinical Infections of *S. enterica*
  - Acute gastroenteritis/Salmonellosis (food poisoning with vomiting and diarrhea)
    - Caused by ingestion of organism in contaminated food
    - Sources of infection:
      - Poultry, milk, eggs, egg products, and pet handling
      - Other foods such as peanut butter, cantaloupe, cereals, dog food, and tomatoes have also been implicated
      - Insufficiently cooked eggs and domestic fowl, such as chicken, turkey, and duck, can result in infection
      - Cooking utensils, such as knives, pans, and cutting boards used in preparing the contaminated meat, can spread the bacteria to other food.
      - Direct transmission from person to person has been reported in institutions

        - Symptoms
            - Appear 8 to 36 hours after ingestion of contaminated food
            - Nausea, vomiting, fever, and chills
            - Watery diarrhea and abdominal pain
        - Most cases of *Salmonella* gastroenteritis (Salmonellosis) are self-limiting. Symptoms usually disappear within a few days, with few or no complications
        - Infection may be more severe in very young children, elderly adults, and patients with other underlying disease
    - Typhoid fever
        - Febrile disease that results from the ingestion of food contaminated with the organisms originating from infected individuals or carriers
        - Caused by *S. typhi*
        - Clinical Course of typhoid fever
            - Ingestion of organism
            - Development of fever, malaise, anorexia, lethargy, myalgia, and a dull headache
            - Reach the small intestine, invade, and penetrate the intestinal mucosa
            - Enter the lymphatic system and are sustained in mesenteric lymph nodes
            - Released in the bloodstream
            - Spreads to spleen, liver, and bone marrow
            - Engulfed by monocytes and grow intracellularly
            - Rereleased into the blood stream
            - Prolonged bacteremia occurs
            - Organisms invade the gallbladder and Peyer's patches of the bowel
            - "Rose spots" around the umbilical region appear at the second week of fever
        - Serious Complications
            - Necrosis in the gallbladder leading to necrotizing cholecystitis
            - Necrosis of the Peyer's patches leading to hemorrhage and perforation of the bowel
            - Pneumonia, thrombophlebitis, meningitis, osteomyelitis, endocarditis, and abscesses

- Carrier State
  - Individuals who recover from infection may harbor the organisms in the gallbladder
  - Carriers will excrete the organisms in their feces
  - Are a significant source of infection for susceptible persons
  - The carrier state may be terminated by antimicrobial therapy or cholecystectomy if gallbladder infection is evident

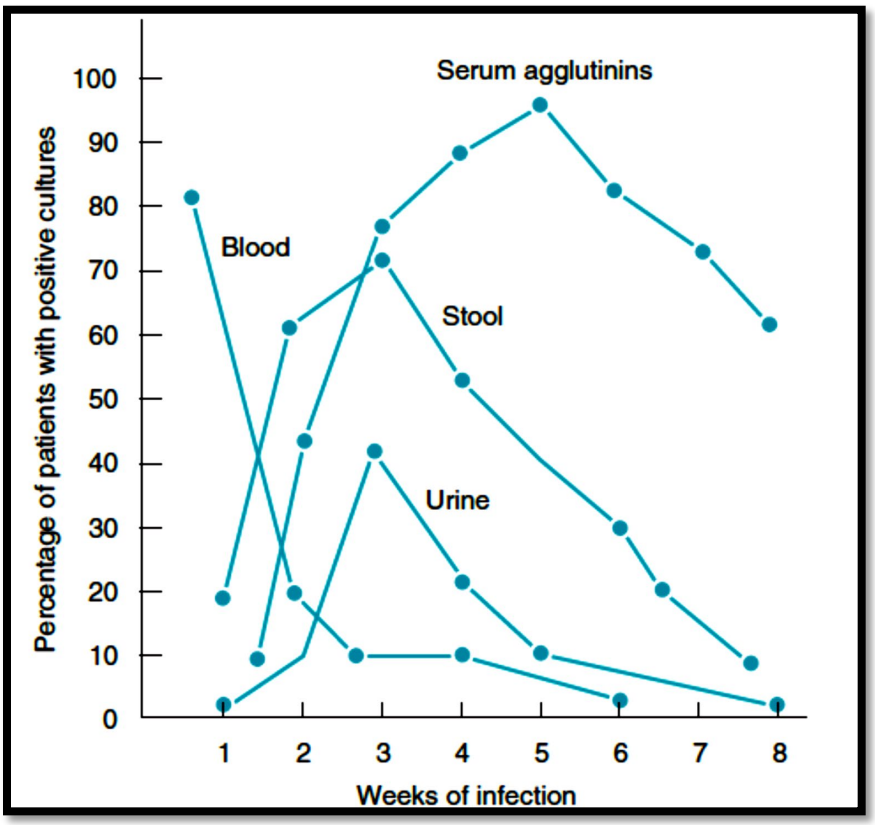


Figure 6.9 – Culture and Serologic Diagnosis of Typhoid Fever. Shows the course of typhoid fever in relations to the number of weeks the infections have been present and the percentage of positive results

- Other enteric fevers
  - Paratyphoid fevers are similar to typhoid fever but are less severe, and the fatality rate is lower
  - Caused by *S. Paratyphi* A, B, C and *S*. Serotype choleraesuis

***Shigella* Species**

- General characteristics of *Shigella* spp.
  - *Shigella* and *Escherichia* are so closely related according to molecular analyses that they should be a single genus
    - For medical purposes, and because of the useful association of the genus epithet with the distinct disease shigellosis or bacillary dysentery, they remain as separate genera
  - Not members of the normal GI flora
  - Genus *Shigella* is named after the Japanese microbiologist Kiyoshi Shiga, who first isolated the organism in 1896
  - Very fragile organism
    - Susceptible to various effects of physical and chemical agents, such as disinfectants and high concentrations of acids and bile

- Antigenic Structures
  - O antigens
    - All *Shigella* spp. possess O antigens
    - Identified by serologic grouping
  - K antigens
    - Certain strains can possess K antigens
      - Can interfere with the detection of the O antigen
    - Heat labile
      - Can be removed by boiling a cell suspension
  - Lack H antigens
    - Non-motile

- Clinical Infections of *Shigella* ssp.
  - All *Shigella* ssp. can cause dysentery; however, species will vary in epidemiology, mortality rate, and severity of disease
  - S. *sonnei* (group D)
    - Most common isolate in the United States
  - *S. flexneri* (group B)
    - Second most common isolated in the United States
    - Affects young children to young adults (appx 25 yrs. old)
  - *S. dysenteriae (group A)*
    - Most common isolate in developing countries
      - Particularly among undernourished children during epidemic outbreaks
    - Most virulent species. Mortality rates range from 5% to 10% and higher
  - *S. boydii* (group C)
    - Also, a common in developing countries

- Clinical infections of *Shigella* spp.
    - Humans are the only known reservoir of *Shigella* spp.
    - Highly communicable
    - Transmission can occur by direct person-to-person contact, and spread can take place via the fecal-oral route
        - May also be transmitted by flies, fingers, and food or water contaminated by infected persons
        - Higher risk in day care centers, crowded populations, and persons performing anal-oral sex
        - Low inoculum: 10 to 200 organisms can cause disease
    - Causes dysentery
        - Local inflammation and shedding of intestinal lining containing mucus, blood, ulcer formation, and tenesmus (rectal prolapse)
        - Infection results from penetration of the mucosal epithelium
            - Production of Shiga toxin produces some of the symptoms
        - Symptoms
            - Appear approximately 24 to 48 hours after ingestion of the organisms
            - High fever, chills, abdominal cramps, and pain
            - Bloody stools containing mucus and numerous leukocytes follow the watery diarrhea, as the organisms invade the colonic tissues and cause an inflammatory reaction
            - Extremely painful bowel movements can lead to tenesmus
        - Usually, non-fatal and self-limiting
            - Life-threatening complication can develop causing extraintestinal complications develop
                - Ileus- an obstruction of the intestines, with marked abdominal dilation, possibly leading to toxic megacolon

- Laboratory Identification of *Shigella* spp.
    - Colony Morphology
        - Initially appear as clear, non–lactose-fermenting colonies
            - Delayed fermentation of lactose on MAC after 48 hours

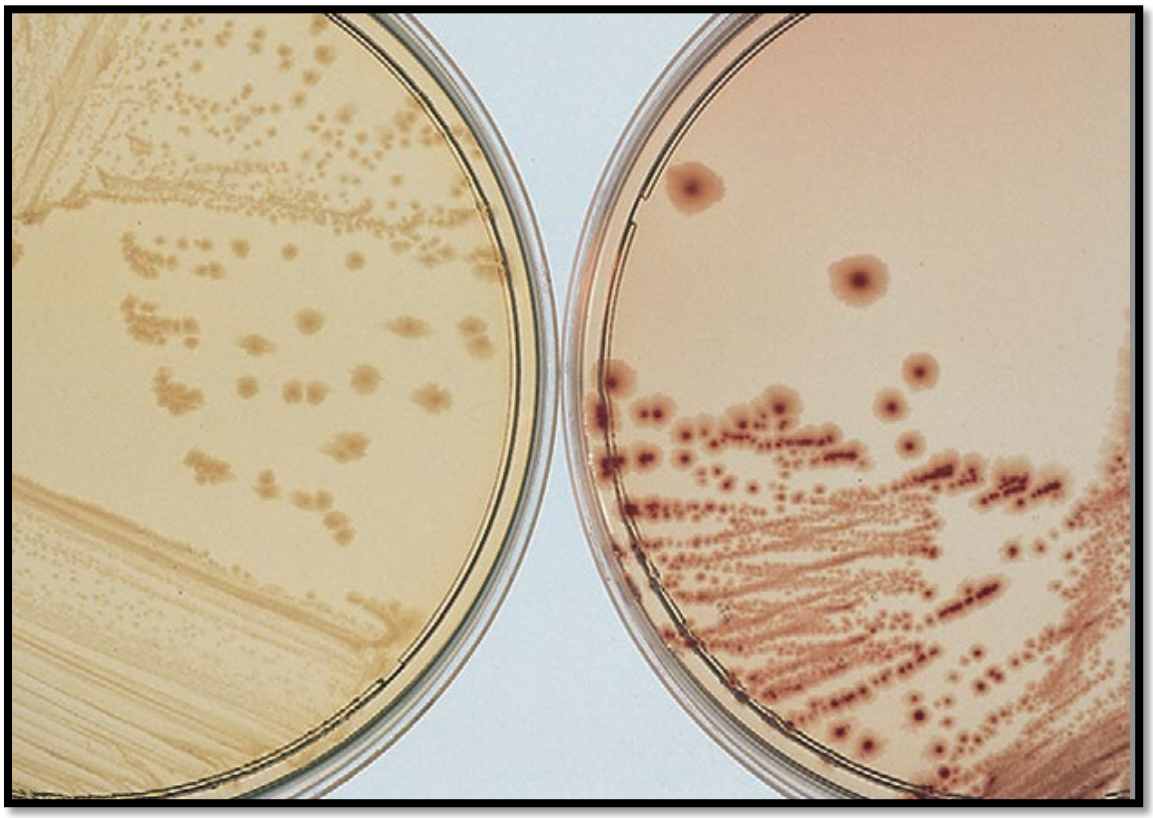


Figure 6.10 – Both are MAC plates with Shigella sonnei growth. (L) Lactose negative appearance at 18-24 hrs. of incubation. (R) Lactose-positive appearance after 48 hrs. of incubation

- Biochemical Identification of *Shigella* spp.
    - IMVIC reaction: - + - -
        - Indole can be variable
    - Generally, they do not produce gas from glucose
    - No urease production
    - No H_2S in TSI
    - No decarboxylation of lysine
    - Cannot utilize acetate or mucate as carbon source
    - ONPG
        - *S. sonnei* is positive for ONPG and ornithine decarboxylase
        - *S. flexneri* is negative for ONPG and ornithine decarboxylase
    - Nonmotile

Table 6.6 – Biochemical and Serologic Differentiation of Shigella Species

Test	*S. dysenteriae*	*S. flexneri*	*S. boydii*	*S. sonnei*
Mannitol fermentation	–	+	+	+
ONPG	V	–	V	+
Ornithine decarboxylase	–	–	–	+
Serogroup	A	B	C	D

From Farmer JJ, et al: Enterobacteriaceae: introduction and identification. In Murray PR, et al, editors: *Manual of clinical microbiology*, ed 9, Washington, DC, 2007, ASM Press.
ONPG, *O*-nitrophenyl-β-D-galactopyranoside; –, ≤9% of strains positive; +, ≥90% of strains positive; *V*, 10% to 89% of strains positive.

Yersinia Species

- General Characteristics
    - 14 named species; most are considered environmental species
        - Three species are considered human pathogens:
            - *Y. pestis*
                - *Y. pestis* is a class A bioterrorism agent
            - *Y. enterocolitica*
            - *Y. pseudotuberculosis*

Yersinia pestis

- Laboratory Identification of *Y. pestis*
    - Colony morphology
        - Grows on routine media such as SBA and MAC agar
        - Grows best at 25° C to 30°C

- Microscopic Morphology of *Y. pestis*
    - Gram-negative, short, plump bacillus
    - Bipolar staining
        - “Safety-pin” appearance when stained with methylene blue

- Clinical infections of *Y. pestis*
    - Causes plague
        - Bubonic, septicemic, and pneumonic forms
        - Bubonic form
            - Transmitted through bite of infected fleas
            - Symptoms
                - Appear 2 to 5 days after infection
                - High fever with painful regional lymph nodes known as buboes (swollen lymph nodes) will appear
        - Septicemic form
            - Occurs when bacteria migrate to the bloodstream
        - Pneumonic plague
            - Transmitted through respiratory droplets
            - Occurs secondary to bubonic plague or the septicemic form when organisms proliferate in the bloodstream and then to the respiratory tract
            - Fatality rate is 100% if untreated

Yersinia enterocolitica

- Infections occur primarily in Europe

- Cases in the northeastern U. S. And Canada have been reported

- Acquired from contact with swine, cats, dogs
    - Ingestion of contaminated food that contacted fecal material
    - Refrigeration is ineffective because it survives cold temperatures

- Clinical infections of *Y. enterocolitica*
    - Acute enteritis
        - Most common manifestation
        - An appendicitis-like syndrome
    - Gastroenteritis
    - Arthritis
        - Extraintestinal form
        - Usually follows a GI episode

- Laboratory Identification of *Y. enterocolitica*
    - Colony morphology
        - Grows on Cefsulodin-irgasan-novobiocin (CIN) agar
            - Selective medium is commonly used to detect the presence of *Y. enterocolitica* in feces
            - Contains cefsulodin, irgasan, novobiocin, bile salts, and crystal violet as inhibitory agents to inhibit the growth of normal colon flora

Yersinia pseudotuberculosis

- A pathogen primarily of rodents, particularly guinea pigs

- Human infections
  - Rare
  - Transmission via close contact with infected animals or their fecal material or ingestion of contaminated drink and foods
  - Causes
    - Mesenteric lymphadenitis
      - Presentation similar to appendicitis
    - Pseudo tubercles
      - A nodule histologically similar to a tuberculous granuloma

- Laboratory Identification *Y. Pseudotuberculosis*
  - Appears like a typical plague bacillus (*Y. Pestis*)
    - Differentiating from *Y. pseudotuberculosis* from *Y. pestis*
      - *Y. pseudotuberculosis* is motile at 18°C to 25°C, produces urease, and ferments rhamnose

Table 6.7 – Differentiation of Selected Species within the Genus Yersinia

Test	*Y. pestis*	*Y. enterocolitica*	*Y. pseudotuberculosis*
Indole	–	d	–
Methyl red	+	+	+
Voges-Proskauer			
25°C	–	d	–
37°C	–	–	–
Motility			
25°C	–	+	+
37°C	–	–	–
β-Galactosidase	+	+	+
Christensen urea	–	+	+
Phenylalanine deaminase	–	–	–
Ornithine decarboxylase	–	+	–
Acid produced from			
Sucrose	–	+	–
Lactose	–	–	–
Rhamnose	–	– or +*	+
Melibiose	–	– or +*	+
Trehalose	–	+ or –	+
Cellobiose	–	+	–

Modified from Washington J: *Laboratory procedures in clinical microbiology*, ed 2, New York, 1985, Springer-Verlag.
+, ≥90% positive reaction within 1 or 2 days; –, no reaction (≥90%) in 30 days; *– or +,* most strains negative, some cultures positive; *+ or (+),* most reactions occur within 1 or 2 days, some are delayed; *d,* different reactions; *+ or –,* most cultures positive, some strains negative.
*Test results at 25°C.

New Genera of Enterobacteriaceae

- *Budivicia*
- *Buttiauxella*
- *Cedecea*
- *Ewingella*
- *Kluyvera*
- *Leclercia*
- *Leminorella*
- *Moellerella*
- *Obesumbacterium*
- *Photorhabdus*
- *Rahnella*
- *Tatumella*
- *Trabulsiella*
- *Yokenella*

Kluyvera

- *Kluyvera* is composed of three closely related species:
    - *K. ascorbata* (the most common clinical isolate),
    - *K. cryocrescens*
    - *K. georgiana*

- Isolated from respiratory, urine, and blood cultures
    - Most strains are nonpigmented, but occasional isolates may produce a reddish-blue or violet pigment on blood containing agar
    - All species resemble *E. coli* colonies growing on MAC agar

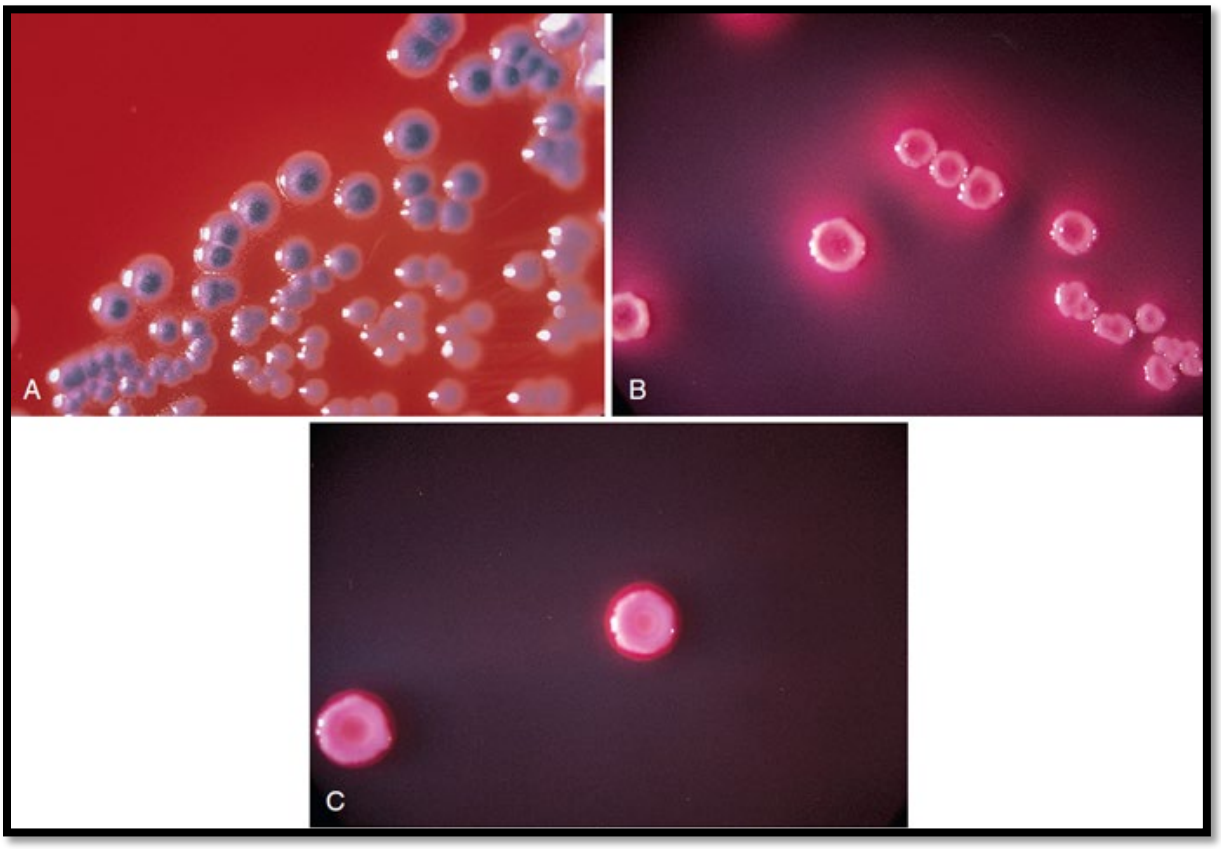


Figure 6.11 – A) *Blue-violet pigment of Kluyvera spp. Growing on sheep blood agar (SBA). This Kluyvera species resembles the colony morphology of Escherichia coli growing on MacConkey (MAC) agar. B. Appearance of K. cryocrescens growing on MAC agar. C. Appearance of K. ascorbata growing on MAC agar*

Topic: Laboratory Diagnosis

Topic Objectives: Discuss laboratory diagnosis of the family Enterobacteriaceae to include specimen processing, isolation and identification, pathogen screening, serologic grouping, and schema

Laboratory considerations for *Enterobacteriaceae*

- Specimen collection and transport
    - Laboratories must provide appropriate transport media, such as Cary-Blair, Amies, or Stuart media
    - Must encourage immediate transport of clinical samples to the laboratory for processing, regardless of the source of the clinical specimen

- Isolation and identification
    - Consider the site of origin
    - Enteric opportunistic organisms isolated from sites that are normally sterile are highly significant
    - In stool cultures, complete identification should be directed only toward true intestinal pathogens

- Direct microscopic examination
    - Not very helpful in differentiating species
        - Characteristics are indistinguishable from other gram-negative bacteria
        - However, smears prepared directly from sterile body fluids or exudates from uncontaminated site can be examined microscopically for the presence of gram-negative bacteria
        - May aid the clinician in the preliminary diagnosis of the infection, and appropriate therapy can be instituted immediately

Screening stool cultures for pathogens

- Separate enteric pathogens from normal enteric flora
    - *Salmonella*, *Shigella*, *Yersinia*, *E. coli* O157:H7, *Aeromonas*, *Campylobacter*, *Vibrio*, and *Plesiomonas shigelloides*
    - All fecal specimens should be routinely screened for *Salmonella*, *Shigella*, and *Campylobacter*
        - Other pathogens should be addressed based on patient history and travel
    - Many laboratories now screen for *E. coli* O157:H7

- **Fecal pathogens are generally non-lactose fermenters (NLFs)**
    - Clear or colorless and translucent colonies on MAC agar
    - Some NLFs such as *Proteus*, *Providencia*, and *Pseudomonas*, as well as delayed lactose-fermenting organisms (e.g., *Serratia* and *Citrobacter*) need to be isolated and screened to rule out a potential pathogen

- In addition to routine media (SBA, MAC), stool samples can also be plated on highly selective media discussed at the beginning of this lesson:
    - Hektoen enteric (HE)
    - Xylose lysine deoxycholate agar (XLD)
    - Eosin-methylene blue (EMB) agar

- Additional selective media used to screen for enteric pathogens, include:
    - Salmonella-Shigella Agar (SS)
    - CHROM agar
    - MacConkey Agar with Sorbitol (SMAC)
        - E. coli O157:H7
    - Cefsulodin-Irgasan-Novobiocin Agar (CIN)



Figure 6.12 – Clear, green colonies of Shigella growing on Hektoen enteric (HE) agar



Figure 6.13 – Clear, green colonies of Shigella growing on H2S producing colonies of Salmonella growing on xylose-lysine-desoxycholate (XLD) agar

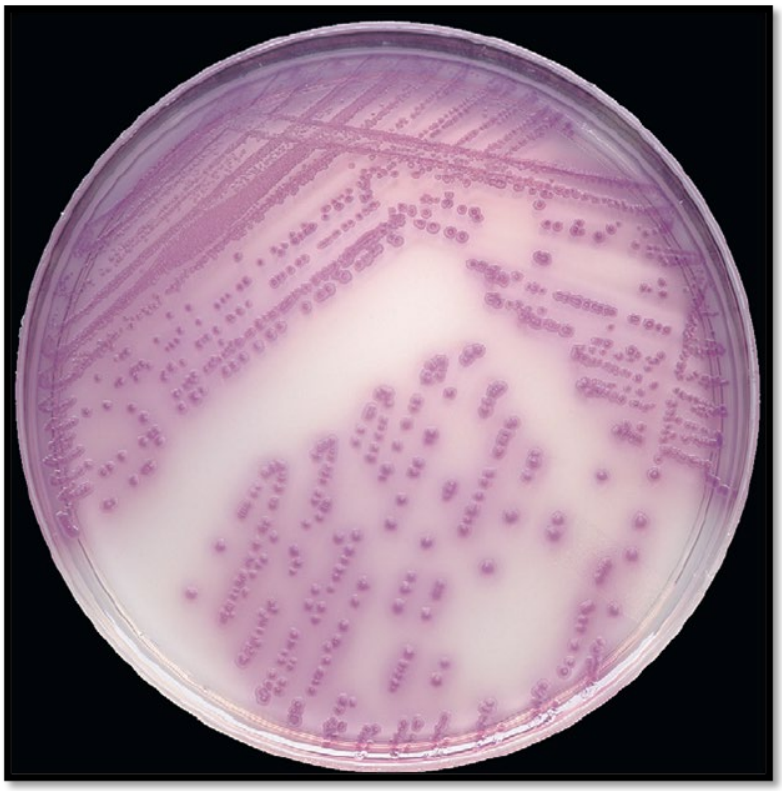


Figure 6.14 – Salmonella growing on Chromagar Salmonella differential agar

Biochemical Identification of Stool Pathogens

- Once isolate has been determined to be a member of the family Enterobacteriaceae, further identification can occur by using:
  - Colony morphology
  - Conventional biochemical tube testing
    - TSI and LIA combination for determine glucose and lactose or sucrose utilization and H_2S production
  - Biochemical testing
    - IMVIC reactions
    - Motility
    - Urea hydrolysis
    - Citrate utilization
    - Carbohydrate fermentation
  - Commercial identification kits
    - API strips, etc.
  - Automated testing
    - VITEK™, etc.
  - Serological testing (grouping)
    - *E. coli* O157:H7
    - *Salmonella*
    - *Shigella*
      - A through D serogroups

Table 6.8 – Lysine Iron Agar (LIA) and TSI Screening

	TSI Reactions							
LIA Reactions	**K/A H_2S**	**K/AG H_2S**	**K/AG**	**K/A**	**A/A H_2S**	**A/AG**	**A/A**	**K/K**
R/A		*P. vulgaris* *P. mirabilis*	*M. morganii* *Providencia*	*M. morganii* *Providencia*	*P. vulgaris* *P. mirabilis*	—	*Providencia*	—
K/K H_2S	*Salmonella** *Edwardsiella*	*Salmonella** *Edwardsiella**	*Salmonella**	*Salmonella**	—	—	—	—
K/K	*Salmonella*	—	*Hafnia* *Klebsiella* *Serratia*	*Salmonella** *Plesiomonas*† *Hafnia*	—	*Klebsiella* *Enterobacter* *E. coli*	*Serratia*	*Pseudomonas*†
K/A H_2S	—	*Salmonella**	—	*Serratia*	—	—	—	—
K/A	—	*Citrobacter*	*Salmonella** *Shigella* *Aeromonas*† *E. coli* *Enterobacter* *Citrobacter*	*Shigella** *Yersinia* *Aeromonas*† *E. coli* *Enterobacter*	*Citrobacter*	*Aeromonas**† *E. coli* *Citrobacter* *Enterobacter*	*Aeromonas**† *Yersinia* *Citrobacter* *Enterobacter*	—

Data from Microbiology Laboratory, The Ohio State University Hospitals and Maureta Ott, Columbus, OH.
A, Acid; *G,* gas; *H_2S,* hydrogen sulfide; *K,* alkaline; *LIA,* Lysine-iron agar; *R,* deamination (red slant); *TSI,* triple sugar iron.
*Results of TSI and LIA reactions in this category indicate a potential pathogen; additional tests must be performed.
†Oxidase-positive.

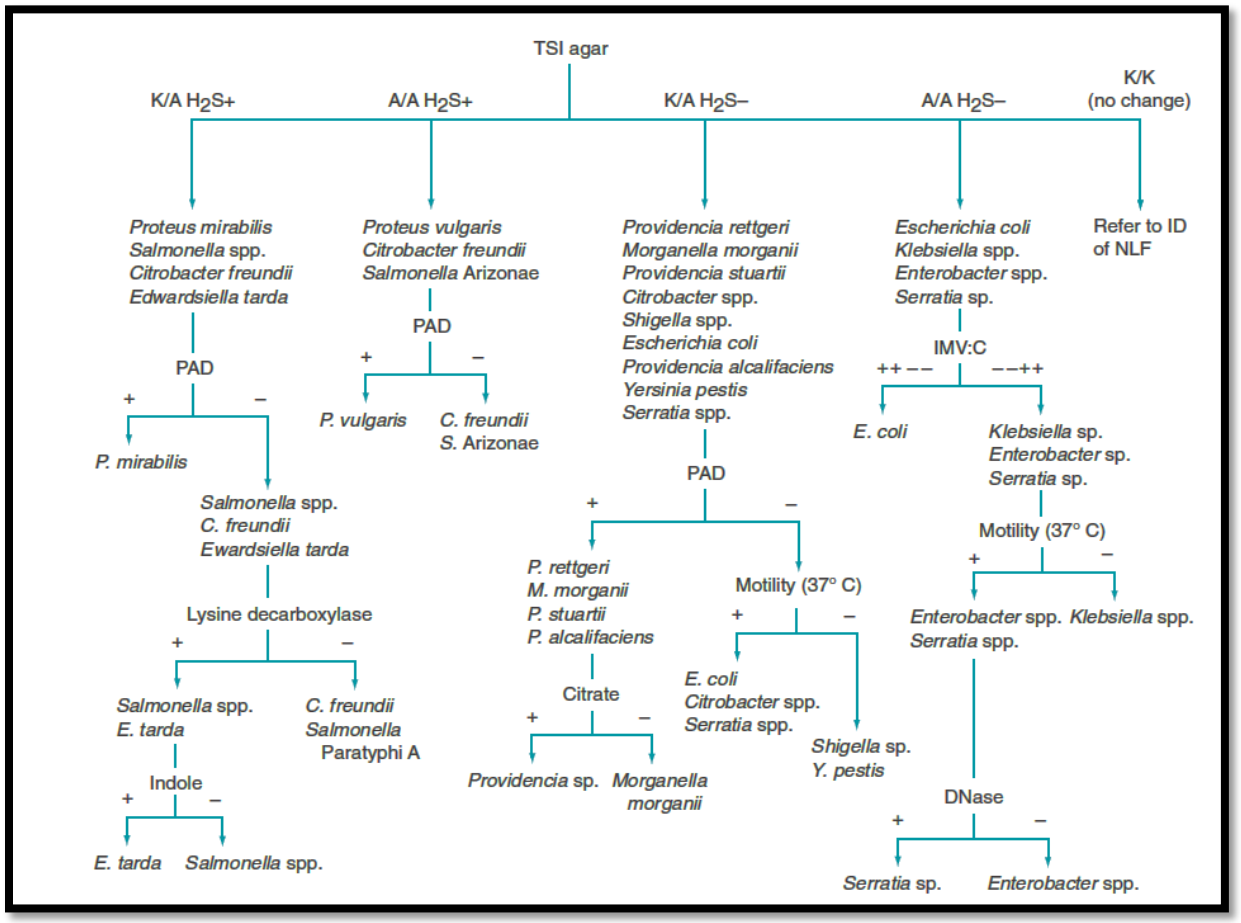


Figure 6.15 – Flow chart for the presumptive identification of commonly encountered Enterobacteriaceae on triple sugar iron (TSI) agar. (A) = acid; (IMV:C) = indole, methyl red, Voges-Proskauer, citrate; (K)= alkaline; (NLF)= non lactose fermenter; (PAD) = phenylalanine deaminase

Checks on Learning

1. Which organism is the most common cause of urinary tract infections (UTIs) in humans?

2. What are the categories of diarrhea-causing *Escherichia coli*?

3. What disease is produced by a *Salmonella* spp.? Is it a severe form of enteric fever?

Lesson: **6.2 Other infectious gram-negative bacilli**

Lesson Objective: **6.2.1 Differentiate other infectious gram-negative bacilli**

Topics:
Vibrio
Aeromonas
Plesiomonas
Campylobacter and Campylobacter like species

Topic Objectives:
Describe the general characteristics of *Vibrio* family to include habitats, clinical manifestations, microscopic morphology, physiology, mode of transmission, pathogenesis, and spectrum of disease
Describe biochemical testing of *Vibrio* family
Discuss antimicrobial susceptibility
Describe the general characteristics of *Aeromonas* to include habitats, microscopic morphology, physiology, and mode of transmission
Describe biochemical testing of *Aeromonas*
Discuss antimicrobial susceptibility of *Aeromonas*
Describe the general characteristics of *Plesiomonas* to include habitats, microscopic morphology, and mode of transmission
Describe the general characteristics of *Campylobacter* and *Campylobacter* like species to include habitats, clinical manifestations, microscopic morphology, physiology, mode of transmission, pathogenesis, and spectrum of disease
Describe biochemical testing of Campylobacter and Campylobacter-like species
Discuss antimicrobial susceptibility of *Campylobacter* and *Campylobacter* like species

Topic: *Vibrio*

Topic Objectives: Describe the general characteristics of *Vibrio* family to include habitats, clinical manifestations, microscopic morphology, physiology, mode of transmission, pathogenesis, and spectrum of disease
Describe biochemical testing of *Vibrio* family
Discuss antimicrobial susceptibility

General Characteristics

- 10 species involved in human infections
- Facultative anaerobic
- Most *Vibrio*s are halophilic
    - Except *V. cholerae* and *V. mimicus*
- Commonly found in a wide variety of aquatic environments, including fresh water, brackish or estuarine water, and marine or salt water
- Temperature sensitive
    - *Vibrio* spp. It can be easily isolated in temperate climates in which water temperatures exceed 20° C, as in the summer months
    - Numbers decline markedly in the winter months
- Usually involved in epidemics
    - Cyclical
- Recent increase (since 2012) in the isolation of *Vibrio* has been reported possibly due to:
    - Increase in ocean water temperature
    - Travel to coastal or cholera endemic areas
        - Inadequate sanitation in developing countries
    - Increased consumption of seafood
        - Undercooked or raw seafood
    - Increased use of recreational water facilities
        - Increased exposure
    - Larger population of immuno-compromised individuals
    - Increased awareness of the existence and clinical significance

Clinical Infections of *Vibrio* spp.

- Low frequency of isolation in non-coastal areas
    - Good medical history is extremely important

- Factors to be considered:
    - Recent consumption of raw seafood (especially oysters)
    - Recent immigration or foreign travel
    - Gastroenteritis with rice-water stools
    - Accidental trauma incurred during contact with fresh, estuarine, or marine water or associated products

Table 6.9 – Clinical Infections Associated with Vibrio Species

Species	Clinical Infection	Frequency
V. cholerae O1	Cholera, gastroenteritis, wound infections, bacteremia	Common
V. cholerae O139	Cholera	Common
V. cholerae non-O1	Gastroenteritis, septicemia, ear infections	Relatively common
Vibrio parahaemolyticus	Gastroenteritis, wound infections	Common
Vibrio vulnificus	Septicemia, wound infections	Common
Vibrio alginolyticus	Wound infections, ear infections, conjunctivitis, respiratory infections, bacteremia	Common
Vibrio mimicus	Gastroenteritis, ear infections	Uncommon
Vibrio fluvialis	Gastroenteritis	Uncommon
Vibrio furnissii	Gastroenteritis	Uncommon
Vibrio cincinnatiensis	Meningitis	Rare
Vibrio metschnikovii	Septicemia, peritonitis	Rare
Vibrio harveyi	Wound infections	Rare
Grimontia hollisae	Gastroenteritis	Uncommon
Photobacterium damselae	Wound infections	Uncommon

From Abbott SL et al: *Vibrio* and related organisms. In Versalovic J et al, editors: *Manual of clinical microbiology,* ed 10, Washington, DC, 2011, ASM Press, p. 666.

Virulence and Antigenic Factors

- Little is known about the antigenic structure of *Vibrio* except for *V. cholerae*, *V. parahaemolyticus*, *V. fluvialis* and *V. vulnificus*

- O antigen
    - Somatic antigen
    - Has six serogroups
        - O1 and 0139 are most important and associated with epidemic cholera

- H antigen - Flagellar antigen

- K antigen - Capsule

Vibrio cholera

- *V. cholera* is the causative agent of cholera
    - A severe diarrheal disease
    - Disease of major public health significance for centuries

- Epidemiology of *V. cholerae*
    - Most epidemics occur in developing countries.
    - Bengal region of India and Bangladesh
    - Cholera is not commonly reported in the United States
        - Most are considered “imported” cases

- Transmission
    - Spread mainly through contaminated water
    - Eating organisms from tainted water
    - Improperly preserved or handled foods

- Clinical Manifestation of cholera
    - Bacteria colonize the small intestine and release toxin, also called choleragen
    - Action of choleragen
        - Toxin causes severe acute gastroenteritis
            - Rice-water stool (watery with flecks of mucus)
            - Vomiting
        - Leads to large loss of water and electrolytes (10-30 movements per day)
        - Dehydration, hypovolemic shock (loss of fluid volume), and metabolic acidosis
        - Can cause death rapidly if fluids are not restored

- Treatment
    - Intravenous (IV) and oral fluids containing electrolytes
    - Antibiotics to reduce organism load
        - Reported resistance to tetracycline and doxycycline

Vibrio parahaemolyticus

- Second most common *Vibrio* species implicated in gastroenteritis after *V. cholerae*

- Discovered in Japan in 1950
    - Large food poisoning outbreak
        - Today called the “Summer diarrhea” of Japan
    - Has been isolated in Europe, the Baltic area, Australia, Africa, Canada, and almost every coastal state in the US

- Limited to coastal areas with 1% to 8% sodium chloride (NaCl)

- Association with at least 30 different marine animal species

- Most cases of gastroenteritis can be traced to recent consumption of improperly cooked raw, or re-contaminated seafood, particularly oysters

- Clinical manifestations
    - Develop after 24 to 48 hours after eating contaminated seafood
        - Self-limiting watery diarrhea and cramping
        - Sometimes vomiting
        - Usually, no fever
    - Extraintestinal wound, eye, ear infections, even pneumonia

Vibrio vulnificus

- Found on the Atlantic, Gulf, and Pacific coasts

- Also known as the “lactose-positive” *Vibrio*

- Acquire infection from raw or undercooked seafood
    - Can cause septicemia
        - Particularly in people with increased serum iron
            - 40% to 60% mortality rate
    - Wound infections
        - Traumatic aquatic wound
            - Presents as cellulitis
        - Can progress to necrotizing fasciitis and/or multiple organ system failure

- In 2007, the CDC required cases of *V. vulnificus* be reported to local and state public health departments

Vibrio alginolyticus

- Rarely isolated and least pathogenic

- Mostly isolated from external sites
    - Eye, ear, wounds, or burns

- Occupational hazard for sailors, fisherman, or those in constant contact with seawater

- Specimen Collection of *Vibrio* ssp.
    - Collect aspirates or tissues
    - Swabs are acceptable if transported in holding medium
    - Do not use buffered glycerol saline

- Culture morphology
    - Blood agar plate (BAP) works well since it contains enough salt for most *Vibrio*s
    - MacConkey (MAC) plates
        - Lactose negative except *V. vulnificus*
    - If clinical history indicates *Vibrio*s, use thiosulfate, citrate, bile salts, sucrose (TCBS)
        - Yellow: *V. Cholerae, V. alginolyticus, V. fluvialis, V. furnissii, V. cincinnatiensis, V. metschnikovii*, and some *V. vulnificus*
        - Green: *V. mimicus, V. parahaemolyticus, P. damselae*, and most *V. vulnificus*

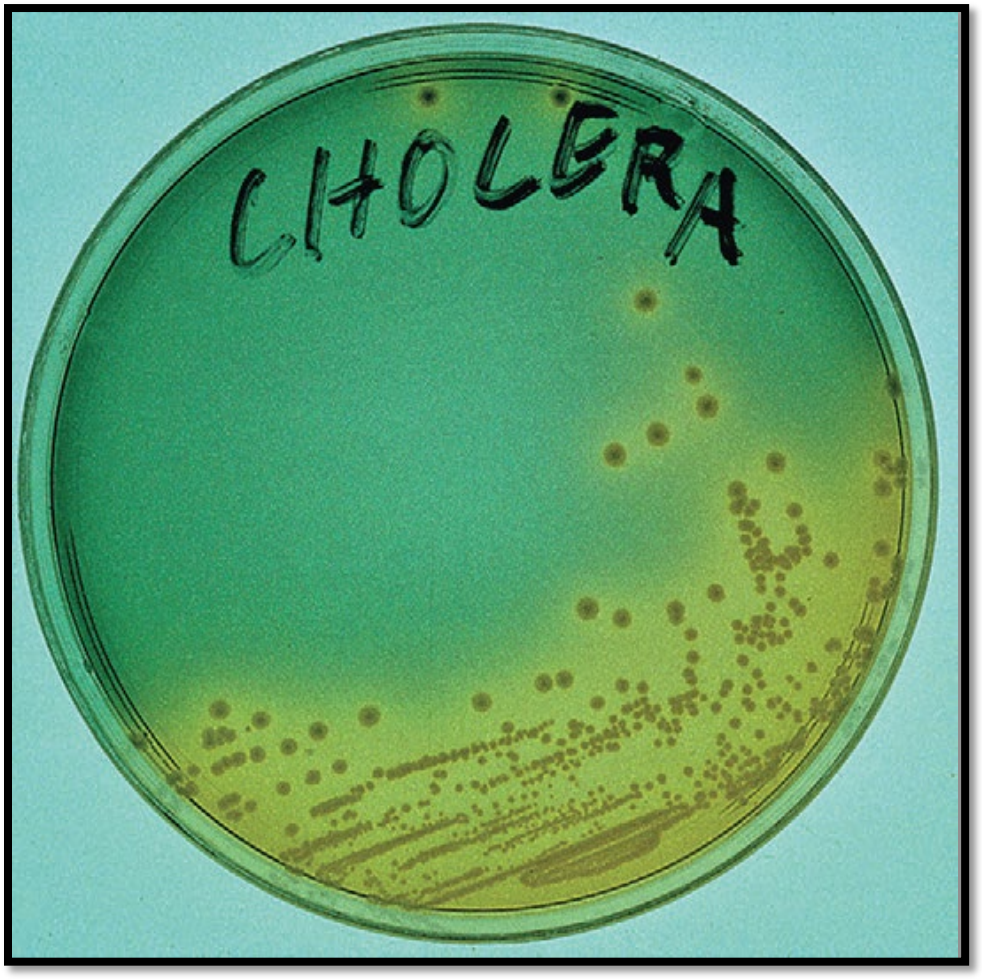


Figure 6.16 – Vibrio cholerae on TCBS

- Microscopic Morphology
    - Small, straight, gram-negative rods
        - 1.4- to 2.6-µm long
        - Can form pleomorphic shapes
            - "Curved" usually only in Gram stains from clinical specimens

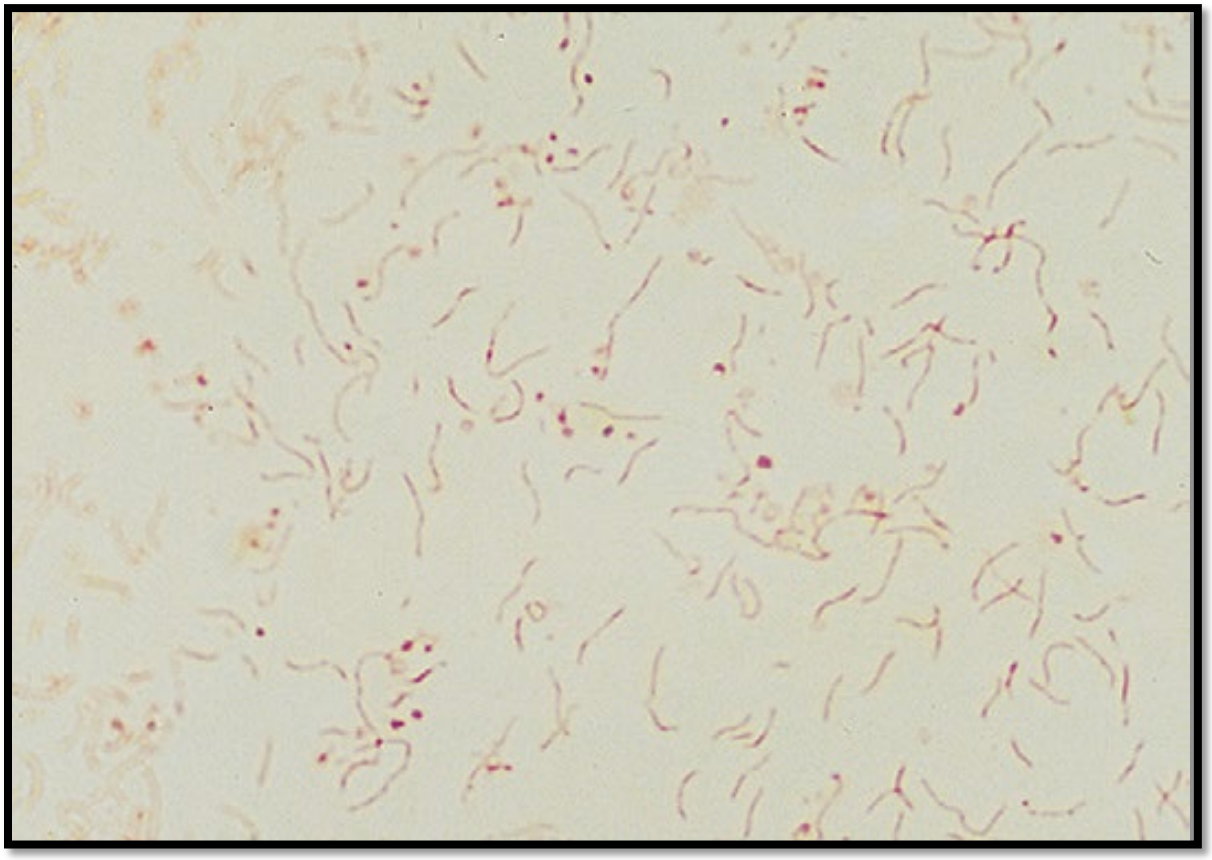


Figure 6.17 – Microscopic morphology of Vibrio spp. On Gram-stained smear

- Biochemical Identification
    - Catalase positive, oxidase positive, ferments glucose and reduces nitrate to nitrite
    - Susceptible to *Vibrio* stat disk
        - *Vibrio* static compound O/129 (2,4-diamino-6,7-diisopropylpteridine)
            - Separates *Vibrios* from *Aeromonas*
        - Isolates from India and Bangladesh
        - Resistance to *Vibrio* stat disks is common in *V. Cholerae* isolates
    - Positive string test
        - Mucoid string seen after emulsifying 2 to 3 colonies in a drop of sodium desoxycholate



Figure 6.18 – O/129 Susceptibility test for Vibrio sp. (L)- Resistant (Aeromonas); (R)- Susceptible (Vibrio)

- Rapid and Semi-automated Identification
  - Most contain databases of the more commonly encountered clinical *Vibrios*
  - Generally, it is inadequate for accurate identification of these species, particularly those that are less commonly encountered
    - Often they are confused with other genera, such as *Aeromonas*
    - Most authorities advocate identification with conventional biochemical
  - If *Vibrios* are identified using a rapid or semi-automated system, ID should be confirmed at reference laboratory or state public health department laboratory

Antimicrobial Susceptibility for *Vibrio* spp.

- Both Mueller-Hinton agar and broth contain sufficient salt to support the growth of *Vibrio* spp.

- Disk or dilution methods
  - CLSI has interpretive guidelines only for *V. Cholerae* and are limited to the following drugs:
    - Ampicillin
    - Tetracyclines
    - Trimethoprim
    - Chloramphenicol

- Most isolates are susceptible to
  - Doxycycline
  - Ciprofloxacin

Topic: *Aeromonas*

Topic Objectives: Describe the general characteristics of *Aeromonas* to include habitats,
microscopic morphology, physiology, and mode of transmission
Describe biochemical testing of *Aeromonas*
Discuss antimicrobial susceptibility of *Aeromonas*

General Characteristics

- Classified in two groups
  - Mesophilic
    - All species motile
    - *A. hydrophila* complex
      - *A. hydrophila, A. bestiarum*, and *A. salmonicida*
    - *A. veronii* complex
      - *A. veronii biovar sobria, A. veronii biovar veronii, A. jandaei, A. trota, A. schubertii*
    - *A. caviae* complex
      - *A. caviae, A. media, A. eucrenophilia*
        - *A. caviae* is most frequently associated with gastrointestinal infections, especially in neonate and pediatric populations
  - Psychrophilic
    - *A. salmonicida*
      - Non-motile, grows best at 22°to 25°C

Clinical Infections of *Aeromonas*

- Gastroenteritis
  - Recognized as enteric pathogens
    - Level, and pattern of virulence, like *E. coli* subgroups
  - Transmitted by:
    - Drinking untreated drinking water
    - Eating contaminated seafood
  - Causes various types of diarrheas:
    - Acute secretory diarrhea with vomiting
    - Acute dysenteric form with blood and mucus
    - Chronic diarrhea lasts more than 10 days
    - Cholera-like disease with rice-water stools
    - Traveler's diarrhea
- Extraintestinal infections
  - Associated with various injuries while working in tainted water
    - Presents as cellulitis
    - Few cases of myonecrosis and necrotizing fasciitis reported
  - *A. hydrophila* is the most isolated in wound infections

- Septicemia
    - Associated with *A. veronii* biovar sobria
    - Occurs in patients that are immuno-compromised, have liver disease, or after a traumatic injury

- Other Infections
    - Osteomyelitis, meningitis, otitis, pelvic abscesses, cystitis, endocarditis, peritonitis, cholecystitis, keratitis, and endophthalmitis

Laboratory Identification of *Aeromonas*

- Culture Morphology
    - Grow readily on most media used for routine and stool cultures
        - Often are overlooked on MAC agar as normal colon flora because most aeromonads ferment lactose
    - On SBA, appear as large, round, raised, opaque colonies with an entire edge and a smooth, often mucoid, surface
    - Pigmentation ranges from translucent and white to buff-colored
    - Hemolysis is variable on SBA, but most major clinical species display strong β-hemolysis
    - On CIN or modified CIN II, appear as a pink-centered colony from the fermentation of mannitol, with an uneven, clear apron

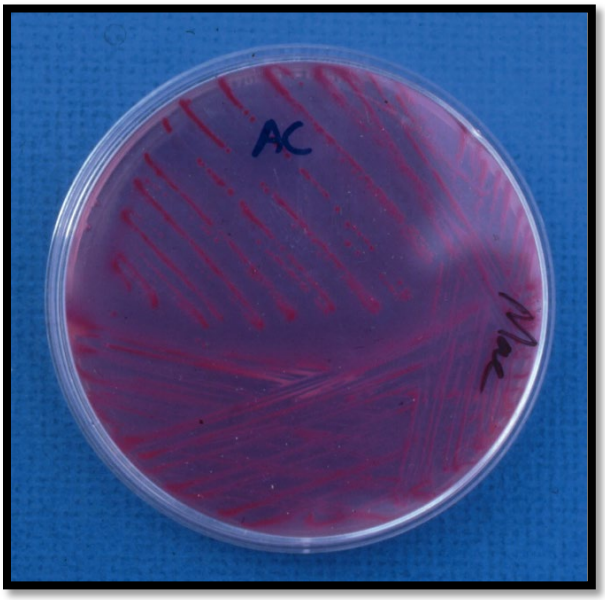


Figure 6.19 – A. caviae lactose fermentation on MAC

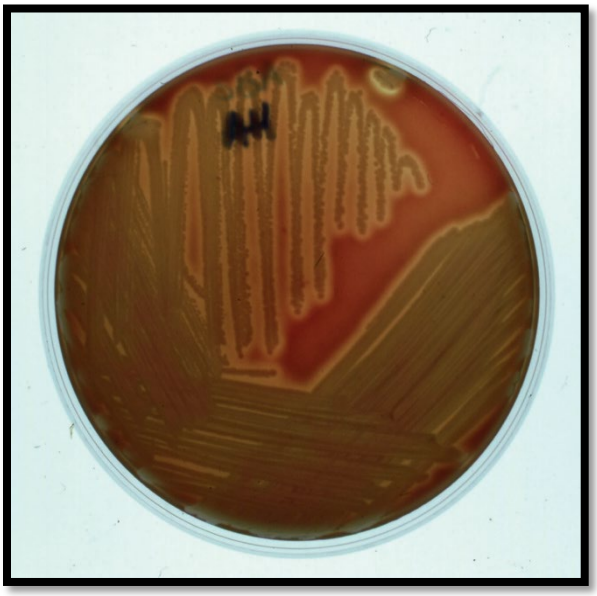


Figure 6.20 – A. hydrophila exhibiting β-hemolysis on SBA

- Microscopic Morphology
    - Gram-negative rods
        - Straight (unlike curved rods of *Vibrio*s)
            - 1.0- to 3.5-µm long

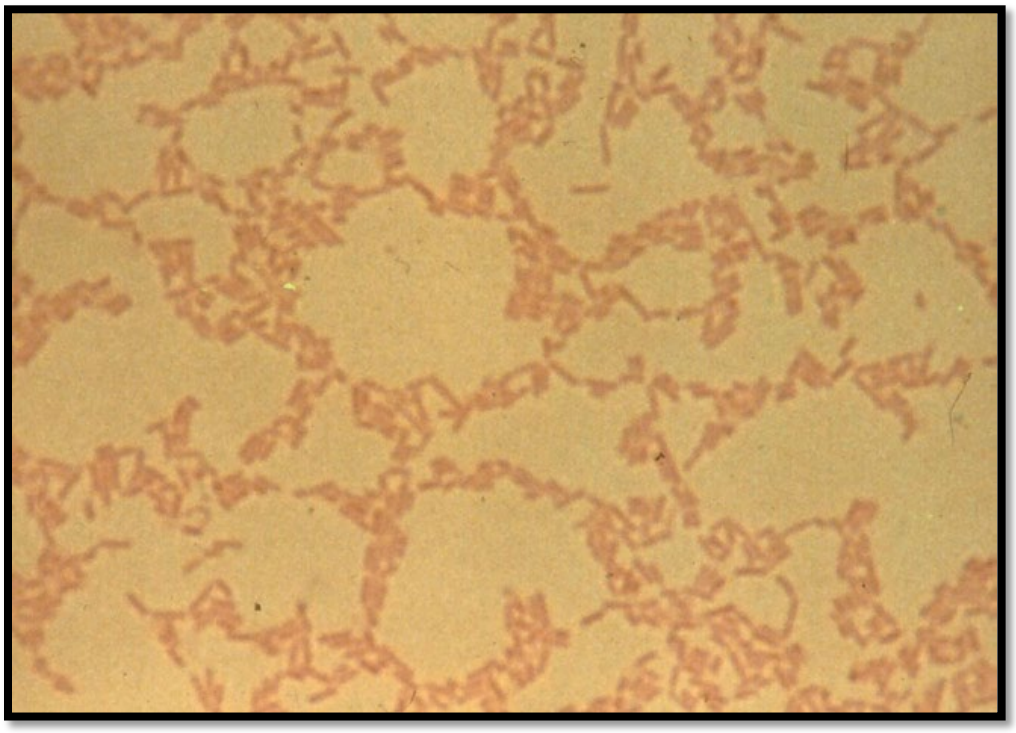


Figure 6.21 – Gram stain of Aeromonas (×1000)

- Biochemical Identification of *Aeromonas*
    - An important screening procedure for aeromonads is to perform an oxidase test and a spot indole test on suspicious colonies on SBA to distinguish from other enterics
        - Oxidase positive
        - Indole positive
            - Most clinically relevant species
    - String test negative
    - Resistant to Vibrio stat disk
    - Grows well in 0% NaCl
    - Inositol negative
    - Definitive identification of the aeromonads is accomplished with a small number of conventional and readily available biochemical tests, including the API 20E strip (BioMérieux)
        - Should always try to identify the species level

Antimicrobial susceptibility of *Aeromonas*

- Antimicrobial susceptibilities should be performed on all clinical aeromonad isolates, if possible

- Per CLSI, susceptibility studies should always be determined by the standard Kirby-Bauer disk diffusion method
    - Due to serious discrepancies in β-lactamase detection by rapid MIC methods that would directly affect the proper treatment of aeromonad infections

- Although most cases of *Aeromonas*-associated gastroenteritis are self-limited, antimicrobial therapy is often indicated

- Always warranted in wound infections and septicemia

- As a genus, aeromonads are almost uniformly resistant to penicillin, ampicillin, and carbenicillin

- Generally susceptible to trimethoprim/sulfamethoxazole (SMZ), aminoglycosides, and quinolone

Table 6.10 – Differential Characteristics for Mesophilic Clinical Aeromonas Species

	Aeromonas Species						
Characteristic	***A. hydrophila***	***A. veronii* biovar sobria**	***A. veronii* biovar veronii**	***A. caviae***	***A. schubertii***	***A. jandaei***	***A. trota***
Esculin hydrolysis	+	–	+	+	–	–	–
Voges-Proskauer	+	+	+	–	V+	–	
Pyrazinamidase activity	V	–	–	+	–	–	–
Arginine dihydrolase	+	+	–	V	+	+	+
Fermentation							
Arabinose	+	–	–	+	–	–	–
Cellobiose	–	–	+	V	–	–	+
Mannitol	+	+	+	+	–	+	+
Sucrose	+	+	+	+	–	–	V
Susceptibility							
Ampicillin	R	R	R	R	R	R	S
Ticarcillin or Piperacilli	V	V	R	V	V	V	S
Cephalothin	V	V	V	V	V	V	V
Decarboxylase							
Lysine	+	+	+	–	+	+	+
Ornithine	–	–	+	–	–	–	–
Indole	+	+	+	V+	–	+	+
H_2S*	+	+	+	–	–	+	+
Glucose (gas)	+	+	+	–	–	+	+
Hemolysis (5% sheep RBCs)	+	+	+	V	+	+	V

From Horneman AJ et al: *Aeromonas*. In Versalovic J et al, editors: *Manual of clinical microbiology,* ed 10, Washington, DC, 2011, ASM Press, p. 658.
+, Positive; –, negative; *R*, resistant; *RBC*, red blood cells; *S*, susceptible; *V*, variable; *V+*, variable usually positive.
*H_2S from gelatin-cysteine-thiosulfate media, not the conventional TSI (triple sugar iron agar) media.

Topic: *Plesiomonas*

Topic Objectives: Describe the general characteristics of *Plesiomonas* to include habitats, microscopic morphology, and mode of transmission

General Characteristics

- *P. shigelloides* is the only species in this genus

- The genus was formerly in the family Vibrionaceae
    - Phylogenetic studies have shown that *Plesiomonas* is actually closely related to members of the family Enterobacteriaceae

- Facultative anaerobe

- Found in soil and aquatic environments
    - Generally found only in the fresh and estuarine waters of tropical and subtropical climates

- Emerged as a potential cause of enteric disease in humans and have also been isolated from a number of extraintestinal infections
    - Underreported because of the similarity to *E. coli* on most ordinary enteric media
    - Antigenically have some similar features to *Shigella* but much lower virulence

Clinical infections of *Plesiomonas*

- Gastroenteritis
    - Two well-known outbreaks in Japan and one in Cameroon
    - Transmission from raw or undercooked seafood or tainted water
    - Three major clinical types of gastroenteritis:
        - Watery diarrhea
        - Subacute or chronic disease
            - Lasts 2 weeks to 3 months
        - Dysenteric form resembling colitis
    - Symptoms
        - 25% to 40% of all patients present with fever vomiting, or both
        - Most common clinical symptom is abdominal pain

- Extraintestinal Infection
    - Vets and zookeepers are more at risk
        - Organism found in warm and cold-blooded animals
            - Dogs, cats, pigs, vultures, snakes, lizards, fish, newts, and shellfish
    - Aqua culturists, fish handlers, and athletes participating in water-related sports are also at risk
    - Bacteremia and meningitis
        - Occur only in severely immuno-compromised patients or neonates

Laboratory Identification of *Plesiomonas*

- Culture Morphology
    - Grow rapidly on most media
        - Forms shiny, opaque, nonhemolytic colonies with slightly raised centers and a smooth and entire edge

- Inositol brilliant green bile agar (IBB) can enhance the isolation of plesiomonads
        - Appear as white or pink
            - Most coliforms will be green or pink

- Biochemical Identification of *Plesiomonas*
    - Oxidase positive
    - Glucose fermenting but does not produce gas
    - Motile
    - Susceptible to *Vibrio* stat disks
    - Positive ornithine and lysine decarboxylases and arginine dihydrolase

- Differentiating *Plesiomonas*
    - Oxidase positive rules out other Enterobacteriaceae
    - Sensitivity to *Vibrio* static agent rules out *Aeromonas*
    - No growth in high salt rules out *Vibrio*
    - Inositol fermentation confirms *Plesiomonas*

Antimicrobial Susceptibility of *Plesiomonas*

- Most cases of plesiomonads gastroenteritis are self-limiting
    - Antimicrobial therapy is indicated for patients with severe or chronic disease

- Extraintestinal infections, particularly among neonates, often require antimicrobial therapy

- General resistance to the penicillin class of antibiotics
    - However, penicillins combined with a β-lactamase inhibitor, as well as trimethoprim-sulfamethoxazole, can be effective

- Reports of resistance to more than one aminoglycoside

- Susceptible to quinolones, cephalosporins, and carbapenems

Table 6.11 – Key Features for the Identification of Vibrio, Aeromonas, and Plesiomonas

Feature	*Vibrio*	*Aeromonas*	*Plesiomonas*
Gram stain reaction	–	–	–
Oxidase activity	+	+	+
O/129* Susceptibility			
150 µg	S	R	S
Growth on TCBS agar	+	–	–
0% NaCl	–/+	+	+
6.5% NaCl	+	–	–
Acid from:			
Glucose	+	+	+
Inositol	–	–	+
Mannitol	+	+/–	–
Sucrose	+/–	+/–	–
Gelatin liquefaction	+	+	–

From Abbott SL et al: *Vibrio* and related organisms. In Versalovic J et al, editors: *Manual of clinical microbiology,* ed 10, Washington, DC, 2011, ASM Press, p. 666.
R, Resistant; S, sensitive; V, variable.
+, Most strains positive; –, most strains negative; +/– or –/+, predominant reaction first.
*Vibriostatic agent (2, 4-diamino-6, 7-diisopropylpteridine).

Topic: *Campylobacter* and *Campylobacter*-like species

Topic Objectives: Describe the general characteristics of *Campylobacter* and *Campylobacter*-like species to include habitats, clinical manifestations, microscopic morphology, physiology, mode of transmission, pathogenesis, and spectrum of disease
Describe biochemical testing of *Campylobacter* and *Campylobacter*-like species
Discuss antimicrobial susceptibility of *Campylobacter* and like species

General Characteristics

- *Campylobacter* and *Campylobacter-like species,* include *Helicobacter* and *Wolinella*
    - *Helicobacter* and *Wolinella* are members of the family Helicobacteraceae

- Many subspecies and biovars
    - *C. jejuni* spp. *jejuni, C. fetus* ssp. *fetus,* and others

- Formerly classified with the *Vibrio*s because of their positive oxidase and characteristic microscopic morphology

- Microaerophilic organisms. Require oxygen, but at a concentration less than that of room air; 5% is normally optimal
    - Environmental requirements make them to appear to be strict anaerobe

- Ranks fourth among the most common causes of foodborne gastrointestinal illness, behind norovirus, *Salmonella*, and *Clostridium perfringens*

Table 6.12 – Clinically Significant Campylobacter Species and Campylobacter-Like Organisms

Species	Clinical Significance
Arcobacter butzleri	Associated with diarrheal disease and bacteremia in humans and in children with recurring gastrointestinal illness (abdominal cramps), endocarditis, and peritonitis
Arcobacter cryaerophilus	Isolated from cases of human bacteremia and diarrhea
Arcobacter skirrowii	Gastroenteritis
Campylobacter jejuni	Most common cause of bacterial diarrhea worldwide
Campylobacter coli	Gastroenteritis, bacteremia in immunocompromised patients
Campylobacter fetus subsp. *fetus*	Bacteremia in immunocompromised patients
Campylobacter concisus	Involved in periodontal disease; has also been recovered from individuals with gastrointestinal illness
Campylobacter curvus	Periodontal disease
Campylobacter gracilis	Periodontal disease
Campylobacter hyointestinalis	Gastroenteritis
Campylobacter lari	Enteritis very similar to that caused by *Campylobacter jejuni,* also septicemia
Campylobacter rectus	Gastroenteritis; associated with periodontal disease and has been recovered from patients with root canal infections and Crohn's disease
Campylobacter sputorum biovar sputorum	Gastroenteritis
Helicobacter pylori	Common cause of duodenal ulcers and type B gastritis; possibly a risk factor in gastric carcinoma
Helicobacter bilis	Bacteremia
Helicobacter canadensis	Gastroenteritis
Helicobacter cinaedi	Colitis and sepsis; recovered from blood of men who have sex with men and feces of children and adult females
Helicobacter felis	Gastritis and ulcers
Helicobacter fennelliae	Proctitis, gastroenteritis, and sepsis, particularly in immunocompromised patients

Clinical Infections of *Campylobacter* and *Campylobacter*-like Species

- The transmission of campylobacteriosis has been attributed to direct contact with animals and handling infected pets, such as dogs, cats, and birds

Campylobacter jejuni

- The most common causes of bacterial gastroenteritis worldwide

- Symptoms
    - Mild abdominal pain within 2 to 10 days after ingestion of the organisms
    - Cramps and bloody diarrhea follow
    - Patients may experience fever and chills and, rarely, nausea and vomiting

- Associated with Guillain-Barre syndrome
    - Patients often test positive for *Campylobacter* antibodies

Campylobacter fetus

- Cause enteric infections in immunocompromised and elderly patients

- Known to cause abortions in cattle, sheep, pigs

Helicobacter pylori

- Colonizes 20% to 40% of adults in the United States

- Major cause of chronic superficial gastritis

- Transmission possible via person-to-person transfer

- Does not invade the tissue but does live in the mucus covering the surface of the stomach
    - Produces urease to breakdown urea to ammonia and neutralize the stomach pH
    - Produces a low-grade inflammatory response
    - Damages tissue, resulting in gastric, peptic, and duodenal ulcers

- Associated with type B gastritis
    - Associated with causing gastric carcinoma (stomach cancer)

Laboratory Identification of *Campylobacter*

- Culture Morphology
  - Campy blood agar plates
    - Brucella agar base with 10% sheep RBCs
  - Skirrow media
    - Blood agar base, lysed horse RBCs
  - Cary-Blair – transport medium for stool
    - Used if there is a delay in processing

- Plates must incubate in a microaerophilic, capnophilic atmosphere at 42°C
  - 5% to 10% oxygen (O_2) and 10% carbon dioxide (CO_2)
  - Incubate 72 hours

- Colony Morphology of *Campylobacter*
  - *C. jejuni*
    - Moist “runny looking” and spreading
    - Non-hemolytic
    - Raised round or flat
  - *C. fetus* subsp. *fetus*
    - Tan or slight pink smooth, convex, translucent colonies

- Microscopic Morphology of Campylobacter
  - Non–spore-forming gram-negative rods
    - May appear as long spirals or ‘S’ or seagull-wing shape

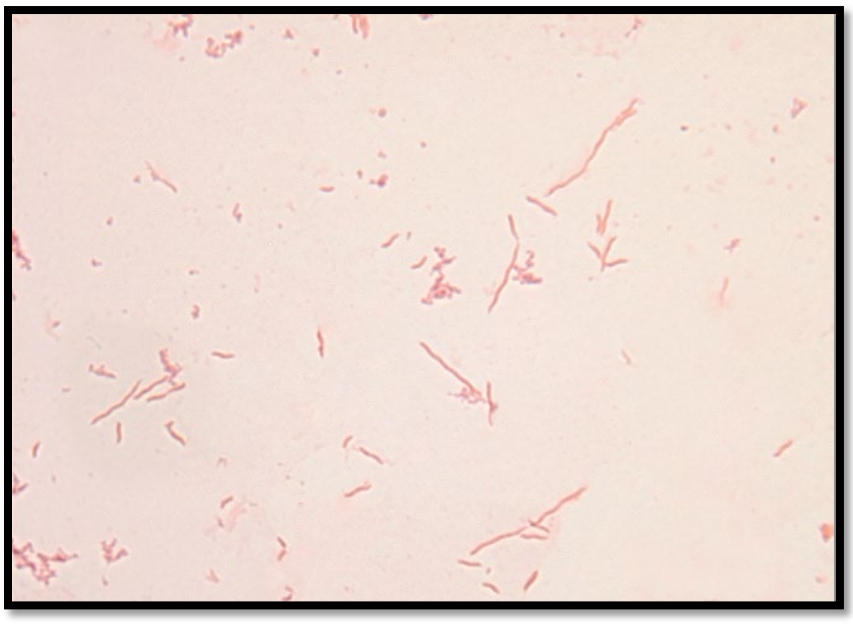


Figure 6.22 – Campylobacter Gram stain

- Biochemical Identification of *Campylobacter*
  - Oxidase positive
  - Motile
    - Exhibits a “darting” motility on hanging drop preparations or when visualized under phase contrast microscopy
      - Organisms should be suspended in Brucella or tryptic soy broth
      - Distilled water and saline inhibit motility
  - Positive hippurate hydrolysis
    - *C. jejuni*

Laboratory Identification of *Helicobacter*

- Usually are identified by non-culture methods

- Gastric biopsy specimen can be tested for the presence of a rapid urease reaction
  - Requires Christensen’s urea medium
  - Rapid color change in 2 hours

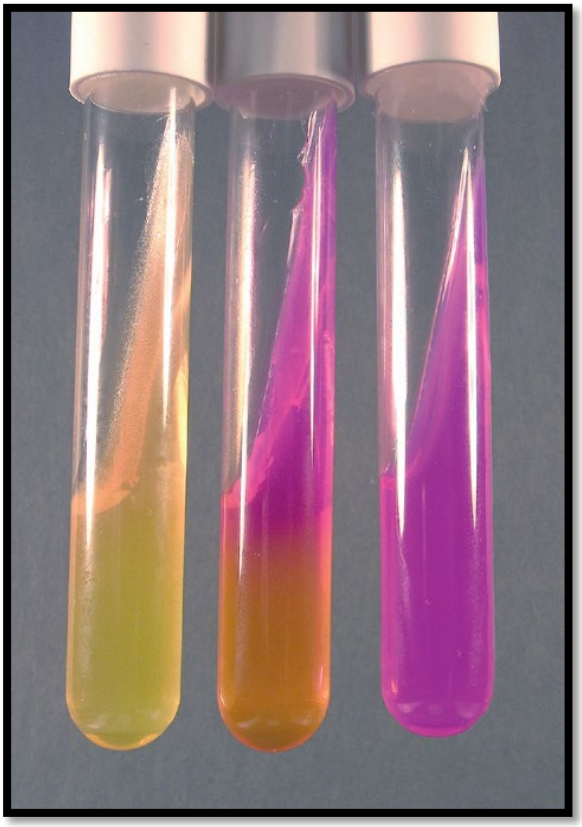


Figure 6.23 – Urease reactions. Left to right, Negative, weak positive, positive

- Other commercial rapid urea tests:
  - Urea breath test
    - Sensitive and specific and is recommended for monitoring therapy
    - Patient is given 13C- or 14C-labeled urea orally

- Biopsy specimen of the stomach
    - Microscopic examination requires Giemsa, Gram, or silver stain

- Immunologic assays
    - Fecal antigen detection
        - DNA amplification tests (e.g., polymerase chain reaction assay)
        - Enzyme-linked immunosorbent assay (ELISA)
        - Immunofluorescence assay (IFA)

Antimicrobial Susceptibility of *Campylobacter* and *Helicobacter*

- Not routinely performed in the clinical microbiology laboratory and is not standardized

- Treatment for *C. jejuni*
    - In most patients, the illness is self-limited and usually resolves in 2 to 6 days
    - Can treat with Azithromycin and erythromycin
    - Gentamicin for systemic infections
    - Untreated patients can remain carriers for several months

- Treatment for *H. pylori*
    - Macrolide, amoxicillin, and a proton pump inhibitor
        - Treat 7 to 14 days

Table 6.13 – Biochemical Tests to Differentiate Campylobacter, Arcobacter, and Helicobacter Species

Species	Catalase	Nitrate Reduction	Urease	H_2S Production (TSI)	Hippurate Hydrolysis	Indoxyl Acetate Hydrolysis	Growth at			Susceptibility to	
							15°C	25°C	42°C	Nalidixic Acid (30 µg)	Cephalothin (30 µg)
Campylobacter jejuni subsp. *jejuni*	+	+	–	–	+	+	–	–	+	+	–
Campylobacter jejuni subsp. *doylei*	V	–	–	–	V	+	–	–	–	+	+
Campylobacter coli	+	+	–	V	–	+	–	–	+	+	–
Campylobacter lari	+	+	–	–	–	–	–	–	+	–	–
Campylobacter fetus subsp. *fetus*	+	+	–	–	–	–	–	+	–	–	+
Campylobacter hyointestinalis	+	+	–	–	–	–	–	+	+	–	+
Campylobacter upsaliensis	–	+	–	–	–	+	–	–	+	+	+
Campylobacter concisus	–	+	–	V	–	–	–	–	+	–	–
Campylobacter curvus	–	+	–	V	V	V	–	–	+	+	ND
Campylobacter rectus	V	+	–	–	–	+	–	–	W	+	ND
Arcobacter butzleri	– V	+	–	–	–	+	+	+	V	V	–
Helicobacter pylori	+	V	–	–	–	–	–	–	V	–	+
Helicobacter fennelliae	+	–	–	–	–	+	–	–	–	+	+
Helicobacter cinaedi	+	+	–	–	–	V	–	–	–	+	+

From Fitzgerald C, Nachamkin I: *Campylobacter* and *Arcobacter;* and Fox JG, Megraud F: *Helicobacter.* In Versalovic J et al, editors: *Manual of clinical microbiology,* ed 10, Washington, DC, 2011, ASM Press, p. 885.
+, Positive result; –, negative result; *ND,* not determined; *TSI,* Triple sugar iron agar slant; *V,* variable result; *W,* weak; *–V*, variable usually negative result.

Checks on learning

1. What is the name of the organism that causes the disease cholera?

2. What species of *Aeromonas* is the most frequently associated with gastrointestinal infections?

3. What organism is the most common cause of bacterial gastroenteritis worldwide?

Lesson: **6.3 Non-fermenting and Miscellaneous Gram-negative Bacilli**

Lesson Objective: **6.3.1 Differentiate non-fermenting and miscellaneous Gram-negative bacilli**

Topics: **General Characteristics**
Clinically significant non-fermentative gram-negative bacilli

Topic Objectives: **Describe the general characteristics of non-fermentative and miscellaneous gram-negative bacilli**
Discuss clinically significant non-fermentative, gram-negative bacilli to include identifying characteristics, clinical infections, biochemical testing of Pseudomonas fluorescent and non-fluorescent group

Topic: General Characteristics

Topic Objectives: Describe the general characteristics of non-fermentative and miscellaneous gram-negative bacilli

General Characteristics

- Most are oxidase positive
    - Most nonfermentive, gram-negative bacilli are oxidase positive, a feature that differentiates them from the Enterobacteriaceae (*Plesiomonas* sp. is the only oxidase-positive member of the Enterobacteriaceae)

- Do not ferment carbohydrates (oxidative-fermentative [O/F] media)

- Grow only in an aerobic environment

- Ubiquitous and found in most environments
    - Soil and water, on plants and decaying vegetation, and in many foodstuffs

- Prefer moist environments
    - Have been isolated in nebulizers, dialysate fluids, saline, catheters, and other devices

- Can easily colonize hospitalized patients, especially those who are immuno-compromised

- Resists treatment with chlorhexidine and quaternary ammonium compounds

- Resistant to multiple classes of antimicrobial agents

- Two categories for energy utilization
    - Oxidizers
        - Oxidize carbohydrates to derive energy for their metabolism
    - Non-oxidizers
        - Asaccharolytic
        - Do not break down carbohydrates at all and are inert or biochemically inactive

- Microscopic Morphology
    - Long, thin, gram-negative bacilli or coccobacilli
        - Motile usually with polar or polar tufts of flagella

Clinical Infections of Non-fermenters

- Account for 15% of all gram-negative bacilli isolated in clinical infections
  - *Pseudomonas aeruginosa* 5-15%
  - *Acinetobacter* spp. are opportunists
    - Accounting for 1% to 3% of all nosocomial infections
  - *Stenotrophomonas maltophilia* is the third most common non-fermentative, gram-negative bacillus isolated in the clinical laboratory

- Septicemia
- Meningitis
- Osteomyelitis
- Wound infections

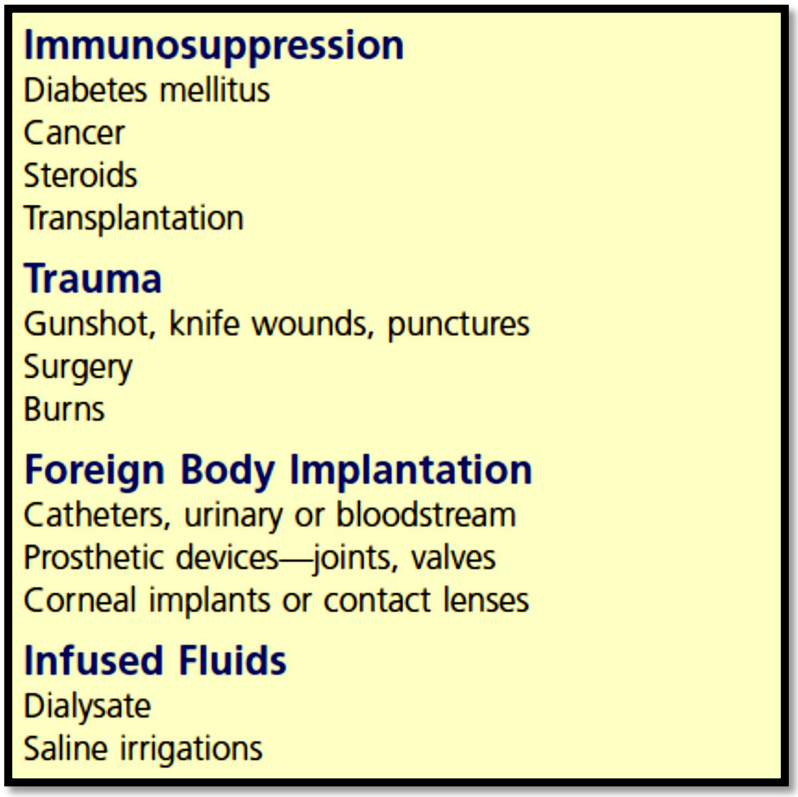


Figure 6.24 – Risk Factors for Diseases Caused by Non-fermentative, Gram-negative Bacilli

Topic: Clinically Significant Non-fermentative, Gram-negative Bacilli

Topic Objective: Discuss clinically significant non-fermentative, gram-negative bacilli to include identifying characteristics, clinical infections, biochemical testing of Pseudomonas fluorescent and non-fluorescent group

***Pseudomonas* Fluorescent Group**

- Members of the *Pseudomonas* fluorescent group include:
  - *P. aeruginosa*
  - *P. fluorescens*
  - *P. putida*
  - *P. veronii*
  - *P. mosselii*
  - *P. monteilii*

Pseudomonas aeruginosa

- Most commonly isolated species of the genus in clinical specimens

- Considered an opportunistic pathogen

- Clinical Infections of *P. aeruginosa*
  - Account for 5% to 15% of all nosocomial infections, especially pneumonia and bacteremia
    - Leading cause of nosocomial respiratory tract infection
  - When *P. aeruginosa* is isolated from a sterile body site, such as blood, pleural fluid, joint fluids or tissues, or cerebrospinal fluid (CSF), it almost always constitutes a true infection
  - Can cause pulmonary disease, especially among individuals with cystic fibrosis
  - Common and less serious conditions associated with *P. aeruginosa*
    - Otitis externa in swimmers or divers
    - Necrotizing skin rash (Jacuzzi or hot tub syndrome)
    - Infections of the nail beds in people with artificial nails
    - Keratitis
    - Endophthalmitis

- Virulence Factors of *P. aeruginosa*
  - Endotoxins
    - Lipopolysaccharide (LPS)
  - Exotoxins
    - Particularly exotoxin A
      - Functions similarly to diphtheria toxin by blocking protein synthesis

- Alginate
  - A polysaccharide polymer
  - Overproduction of this factor produces mucoid colonies
    - Seen primarily in cystic fibrosis patients
- Other virulence factors
  - Motility
  - Pili
  - Proteases
  - Hemolysin
  - Capsule
  - Lecithinase
  - Elastase
  - Dnase

- Laboratory Identification of *P. aeruginosa*
  - Culture and Colony Morphology
    - Most isolates of *P. aeruginosa* are β-hemolytic on sheep blood agar SBA and MAC
    - Produce flat spreading colonies with a characteristic metallic green sheen
    - Many strains also produce a fruity, grapelike odor caused by the presence of 2-aminoacetophenone
    - Produce pyoverdine and pyocyanin
    - When pyocyanin combines with pyoverdine, the green color characteristic of *P. aeruginosa* colonies is produced

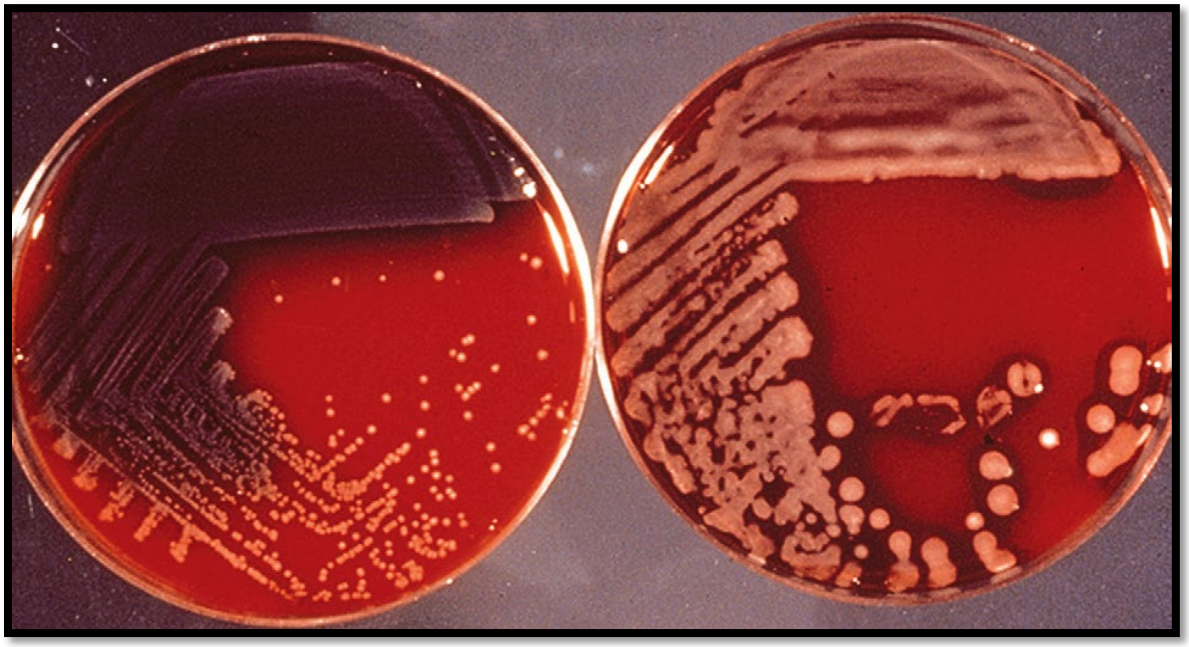


Figure 6.25 – Pseudomonas aeruginosa on sheep blood agar. (L)-Nonmucoid colonies. (R)- Mucoid colonies. Note discoloration of media, especially on the left

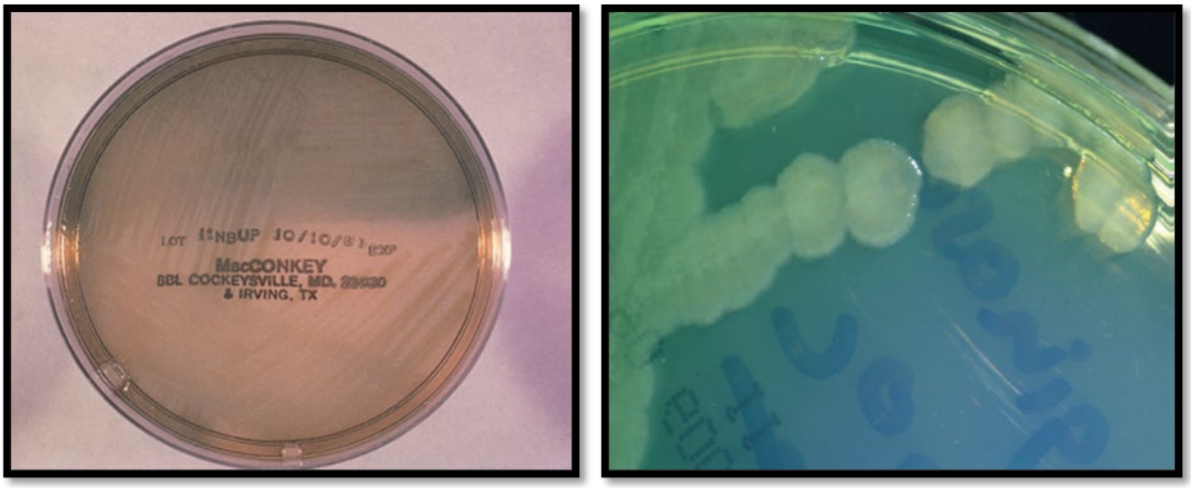


Figure 6.26 – (L) *P. aeruginosa* on MAC agar and (R) Mueller-Hinton agar. Note the blue-green pigment

- Biochemical Identification of *P. aeruginosa*
  - Oxidase positive
  - Motile
  - Indole negative
  - Denitrification of nitrates and nitrites to nitrogen gas
  - Arginine dihydrolase (ADH)–positive
  - Growth at 42°C
  - Citrate-positive
  - Acetamide utilization

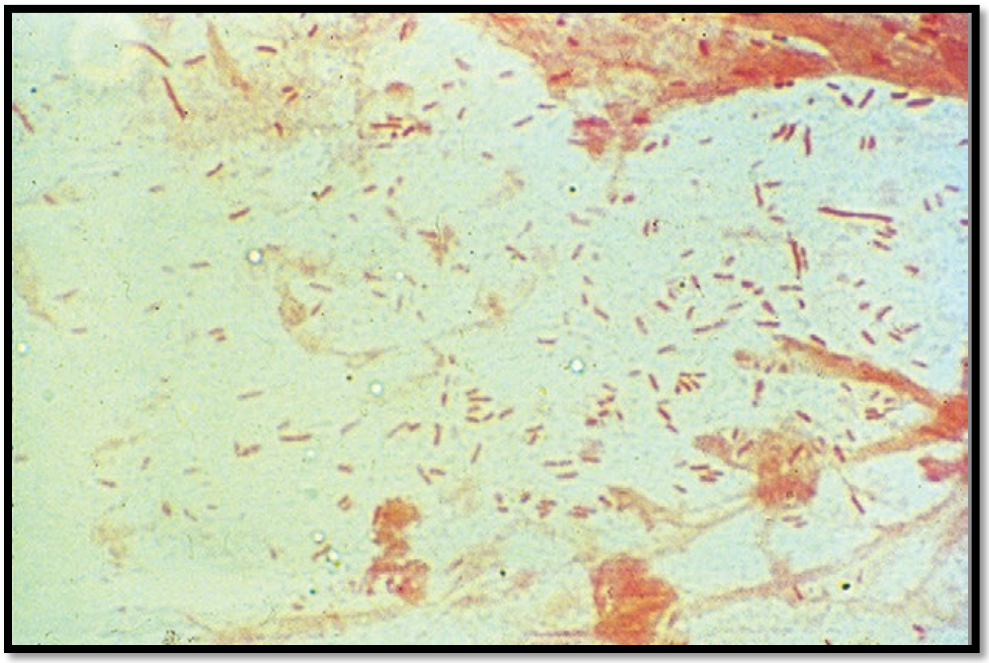


Figure 6.27 – Gram stain of bronchial specimen positive for P. aeruginosa

- Antimicrobial Susceptibility for *P. aeruginosa*
    - Innately resistant to many antimicrobial agents, including:
        - Penicillin
        - Ampicillin
        - First- and second-generation cephalosporins
        - Trimethoprim-sulfamethoxazole (SXT)
        - Chloramphenicol
        - Tetracyclines
    - Susceptible to:
        - Aminoglycosides
        - Semisynthetic penicillins such as piperacillin and ticarcillin
        - Third- and fourth generation cephalosporins
        - Carbapenems (except ertapenem)
        - Fluoroquinolones
    - Incidence of resistance is much higher in nosocomial strains
    - Severe infections usually require combination therapy

Pseudomonas fluorescens* and *Pseudomonas putida

- Low virulence
    - Isolated from respiratory specimens, transfusion-associated septicemia, urine, cosmetics, hospital equipment, and fluids

- Clinical Infections of *P. fluorescens* and *P. putida*
    - UTIs
    - Postsurgical abscesses
    - Empyema
    - Septic arthritis
    - Wound infections

- Laboratory Identification of *P. fluorescens* and *P. putida*
    - Produce pyoverdine, but neither produces pyocyanin
    - Does not grow at 42°C
    - Cannot reduce nitrate to nitrogen gas
    - Can produce acid from xylose
    - Gelatin hydrolysis
        - *P. putida* is negative
        - *P. fluorescens* is positive

- Antimicrobial Susceptibility for *P. fluorescens and P. putida*
    - Susceptible to aminoglycosides, polymyxin, and piperacillin
    - Resistant to carbenicillin and trimethoprim/sulfamethoxazole (SMZ)

***Pseudomonas* Non-fluorescent Group**

- Members of the *Pseudomonas* non-fluorescent group include:
  - *P. stutzeri*
  - *P. mendocina*
  - *P. pseudoalcaligenes*
  - *P. alcaligenes*
  - *P. luteola*
  - *P. oryzihabitans*

Pseudomonas stutzeri

- A rare isolate and even rarer pathogen in the clinical laboratory

- Clinical Infections of P. *stutzeri*
  - Septicemia and meningitis in immunocompromised patients
  - Pneumonia (especially in CF patients)
  - Endocarditis
  - Postsurgical wound infections
  - Septic arthritis
  - Conjunctivitis
  - UTIs

- Laboratory Identification of *P. stutzeri*
  - Culture and Colony Morphology
    - Easily recognizable
      - Wrinkled, leathery, adherent colonies that may produce a light yellow or brown pigment



Figure 6.28 – P. stutzeri on SBA. Note the wrinkled appearance of the colonies

- Biochemical Identification of *P. stutzeri*
  - Arginine dihydrolase negative
  - Starch hydrolysis positive
  - Can grow in an anaerobic environment in nitrate-containing media, producing nitrogen gas

- Antimicrobial Susceptibility for *P. stutzeri*
  - Adherent nature of colonies of *P. stutzeri*, identification of the organism and in vitro susceptibility testing can be unreliable
  - Susceptible to the aminoglycosides, SXT, ampicillin polymyxin, tetracyclines, fluoroquinolones, and third generation cephalosporins
  - Resistant to chloramphenicol and the first- and second-generation cephalosporins

Pseudomonas mendocina

- Considered a contaminant

Pseudomonas pseudoalcaligenes

- Arginine dihydrolase positive and will weakly ferment fructose

- Considered a contaminant

Pseudomonas alcaligenes

- Considered a contaminant

Pseudomonas luteola* and *Pseudomonas oryzihabitans

- Rarely isolated from humans but have been isolated from wounds, abscesses, blood cultures, peritoneal and chronic ambulatory peritoneal dialysis (CAPD) fluids, and other sources

- Gram-negative, non-fermentative, oxidase-negative bacilli. They are Catalase positive and motile

- Oxidize glucose

- Grow on MAC agar, and often produce an intracellular non-diffusible yellow pigment on SBA

- *P. luteola* can be differentiated from *P. oryzihabitans* by ortho-Nitrophenyl-β-galactoside (ONPG) and esculin hydrolysis

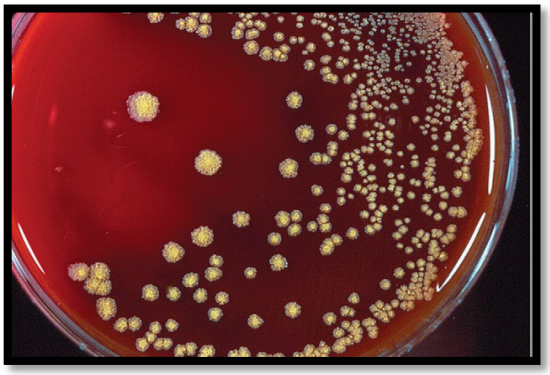


Figure 6.29 – P. oryzihabitans wrinkled yellow colonies on SBA at 48 hours

***Acinetobacter* Species**

- General Characteristics
  - Now a member of the family Moraxellaceae
  - The species most commonly seen in clinical specimens are:
    - *A. baumannii*
    - *A. lwoffii*
    - *A. haemolyticus*
  - Ubiquitous in the environment in soil, water, and food
  - Have been isolated in ventilators, humidifiers, catheters, and other hospital devices
  - 25% of adults carry the organisms on their skin, and about 7% carry the organism in their pharynx
    - 45% of tracheostomy patients may be colonized
    - Were once considered insignificant contaminants

- Clinical Infections of *Acinetobacter* spp.
  - UTIs
  - Pneumonia
  - Endocarditis
    - 22% mortality
  - Septicemia
  - Meningitis
  - Wound and burn infections
  - Eye infections

- Laboratory Identification of *Acinetobacter* spp.
  - Culture and Colony Morphology
    - Have few growth requirements and thus are capable of growing on most laboratory media
    - On MAC:
      - Produce a purplish hue that resembles a lactose-fermenter
    - Strictly aerobic

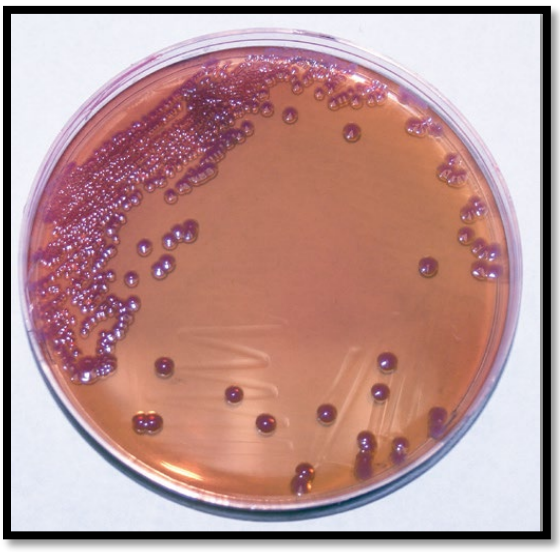


Figure 6.30 – A. baumannii on MAC agar. This organism is saccharolytic, which may cause it to resemble a lactose-fermenting organism

- Microscopic Morphology of *Acinetobacter* spp.
    - Gram-negative coccobacilli or gram-negative cocci
        - Can appear as a gram-positive cocci

- Biochemical Identification of *Acinetobacter* spp.
    - Oxidase negative
    - Catalase positive
    - Non-motile
    - DNA hybridization testing: helpful in grouping
    - *A. baumannii* is saccharolytic and glucose-oxidizing
    - *A. lwoffii* is asaccharolytic and glucose-negative
        - Strictly aerobic
        - Appears as gram-negative coccobacilli

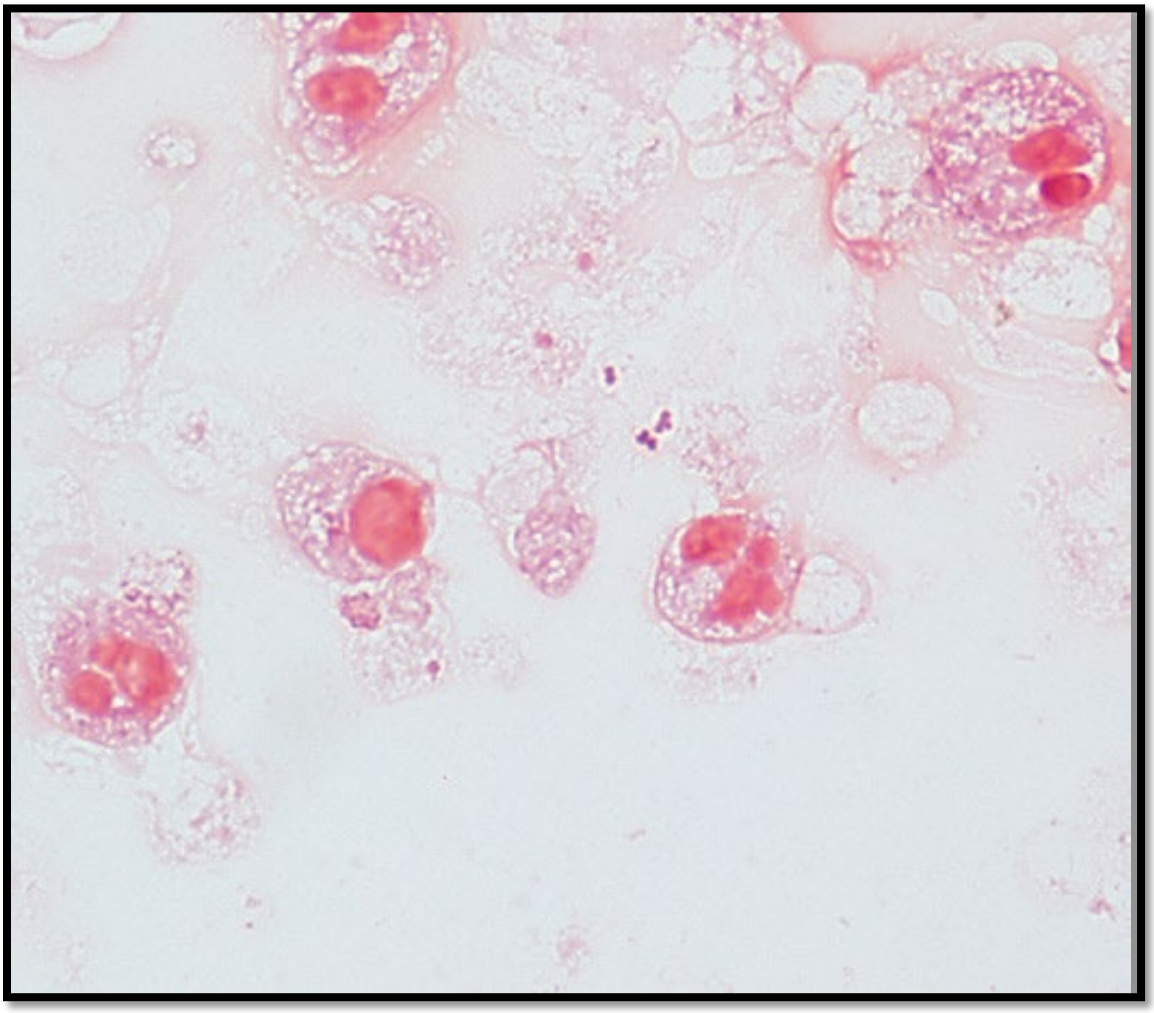


Figure 6.31 – A. baumannii on Gram stain. Note the coccoid nature of the cells as well as the pleomorphic forms

- Antimicrobial Susceptibility of *Acinetobacter* spp.
    - *A. baumannii*
        - Resistant to many antimicrobials, including penicillins, first- and second-generation cephalosporins, and fluoroquinolones
        - Variable susceptibility to the aminoglycosides and β-lactam plus β-lactamase inhibitor combinations (e.g., ampicillin sulbactam, piperacillin-tazobactam)
        - Some strains exhibit resistance to carbapenems; Carbapenemases have been reported
    - *A. lwoffii* - Susceptible to almost all antimicrobials

Stenotrophomonas maltophilia

- **General Characteristics**
    - The third most common non-fermentative, gram-negative bacillus isolated in the clinical laboratory
    - Isolates are ubiquitous in the environment, being found in water, sewage, and plant materials
    - Common contaminate of blood drawing equipment, disinfectants, transducers, and other equipment
    - Not considered part of the normal human microbiota, it can quickly colonize the respiratory tract of hospitalized patients

- **Clinical Infections of *S. maltophilia***
    - Endocarditis
        - Especially in a setting of prior intravenous (IV) drug abuse or heart surgery
    - Wound infections
        - Cellulitis and ecthyma gangrenosum
    - Bacteremia
    - Meningitis
        - Rare
    - UTIs
    - Rarely associated with lower respiratory tract infections, although it has been isolated from 6.4% to 10.2% of patients with CF

- **Laboratory Identification of *S. maltophilia***
    - Culture and Colony Morphology
        - Non-fermentative
        - Colonies may appear bluish on MAC agar
    - Microscopic Morphology of *S. maltophilia*
        - Gram-negative bacillus

- Biochemical Identification of *S. maltophilia*
    - Catalase positive
    - Oxidase negative
    - Dnase positive
    - Esculin positive
    - Lysine decarboxylase positive
    - Gelatin hydrolysis positive

- Antimicrobial Susceptibility of *S. maltophilia*
    - CLSI recommends broth microdilution testing
        - An E-test or agar dilution can be performed
    - Usually susceptible to SXT
        - Drug of choice for most infections
    - May demonstrate in vitro susceptibility include ticarcillin-clavulanate, the fluoroquinolone levofloxacin, and tetracyclines, including tigecycline

Burkholderia cepacia

- General Characteristics
    - A complex of at least 17 distinct genomic species (genomovars)
    - Plant pathogens that have arisen as opportunistic organism usually associated with pneumonia in patients with cystic fibrosis or chronic granulomatous disease (CGD)
    - Isolated from irrigation fluids, anesthetics, nebulizers, detergents, and disinfectants

- Clinical Infections of *B. cepacia*
    - Pneumonia
        - Mainly in patients with CF or chronic granulomatous disease (CGD)
            - Severe with a high mortality rate
    - Endocarditis
        - Specifically in IV drug abusers
    - UTIs
    - Osteomyelitis
    - Dermatitis
    - Wound infections

- **Laboratory Identification of *Burkholderia cepacia***
    - Culture and Colony Morphology
        - Grow well on most laboratory media but may lose viability on SBA in 3 to 4 days
        - Selective media for recovery of *B. cepacia include:*
            - PC (*Pseudomonas cepacia*)
            - OFPBL (oxidative-fermentative base polymyxin B, bacitracin, lactose)

  - BCSA (*B. cepacia* selective agar)
    - Most effective in reducing overgrowth while maintaining good recovery of *B. cepacian*
- Colonies produce a yellow pigment and are non-wrinkled,
- Can produce a non-fluorescing yellow or green pigment that may diffuse into the media



Figure 6.32 – B. cepacia on selective media. (L)- Colonies of B. cepacia on oxidative-fermentative base, polymyxin B, bacitracin, and lactose (OFPBL) agars are yellow. (R)- Colonies of B. cepacia on PC agar are white, surrounded by a pink-red pigment

- Biochemical Identification of *B. cepacia*
    - Oxidase negative
        - May produce a weak, slow, positive oxidase reaction
    - Motility is positive
    - Oxidizes glucose, maltose, lactose, mannitol
    - Lysine decarboxylase (LDC) positive
    - Ornithine decarboxylase (ODC) positive
    - Arginine dihydrolase (ADH) negative

- Antimicrobial Susceptibility *of B. cepacia*
    - Intrinsically resistant to aminoglycosides and polymyxins
        - Many strains are resistant to several β-lactam antibiotics
    - Susceptible to chloramphenicol, ceftazidime, piperacillin, minocycline, some fluoroquinolones, and SXT
    - Susceptibility to the carbapenems is variable
    - Resistance can develop rapidly while a patient is being treated

Burkholderia gladioli

- Resembles *B. cepacia*

- Can be found in CF and chronic granulomatous disease (CGD) patients

- Has been recovered from the blood and tissue of immunocompromised patients, particularly CF patients who have had a lung transplant
- High mortality rate

Burkholderia mallei

- Causes glanders
    - A respiratory tract zoonosis
    - Rare in humans, affects livestock such as horses, mules, and donkeys
        - Can cause severe suppurative or acute pulmonary infections
        - Only one case in 50 years (laboratory accident)
    - Considered by CDC as possible agent of bioterrorism

Burkholderia pseudomallei

- Causes melioidosis
  - An aggressive granulomatous pulmonary disease
  - May cause overwhelming septicemia

- Transmission occurs via ingestion, inhalation, or inoculation of the organisms
  - The organisms are found in water and muddy soils in Southeast Asia (including Vietnam and Thailand), Northern Australia, and Mexico



Figure 6.33 – B. pseudomallei on SBA

- Ashdown medium
  - Wrinkled, deep pink, with earthly odor

Less Commonly Encountered Non-fermentative, Gram-negative Bacilli

- *Moraxella* spp.

- *Alcaligenes*

- *Achromobacter*

***Moraxella* spp.**

- General Characteristics
  - Opportunistic, resides on the mucous membranes of humans
    - Respiratory tract
    - Urinary tract
    - Eyes
  - Asaccharolytic

- Biochemical Identification of *Moraxella* spp.
  - Oxidase positive
  - Non-motile
  - Aerobic
  - Usually susceptible to Penicillin

Alcaligenes* and *Achromobacter

- General Characteristics
  - The family Alcaligenaceae includes the clinically relevant genera of:
    - *Alcaligenes*
    - *Achromobacter*
    - Family Alcaligenaceae also includes the genera *Bordetella*, *Advenella*, *Kertersia*
  - *Alcaligenes* and *Achromobacter* are found in water
  - Resistant to disinfectants, such as chlorhexidine and quaternary ammonium compounds

- Clinical Infections of *Alcaligenes* spp. and *Achromobacter* spp.
  - Isolated from hospitalized patients in urine, stool, sputum, wounds, blood
  - *Alcaligenes faecalis* has been isolated from the blood of patients with and without septicemia
  - *A. xylosoxidans* can cause:
    - Nosocomial infection associated with the use of IV catheters; patients of advanced age and neutropenia are most susceptible
      - Serious illness with 15% fatality rate
    - Otitis media
    - Meningitis
    - Pneumonia
    - Surgical wound infections
    - UTIs
    - Peritonitis
    - Bacteremia
    - Frequent colonizer in the CF patients

- Laboratory Identification of *Alcaligenes* spp*. And Achromobacter* spp*.*
  - Culture and Colony Morphology
    - Obligately aerobic
    - Usually grows well on most laboratory media, including MAC
    - On SBA
      - Most species are non-pigmented
      - Can cause a green discoloration on SBA
    - Some strains may produce a fruity

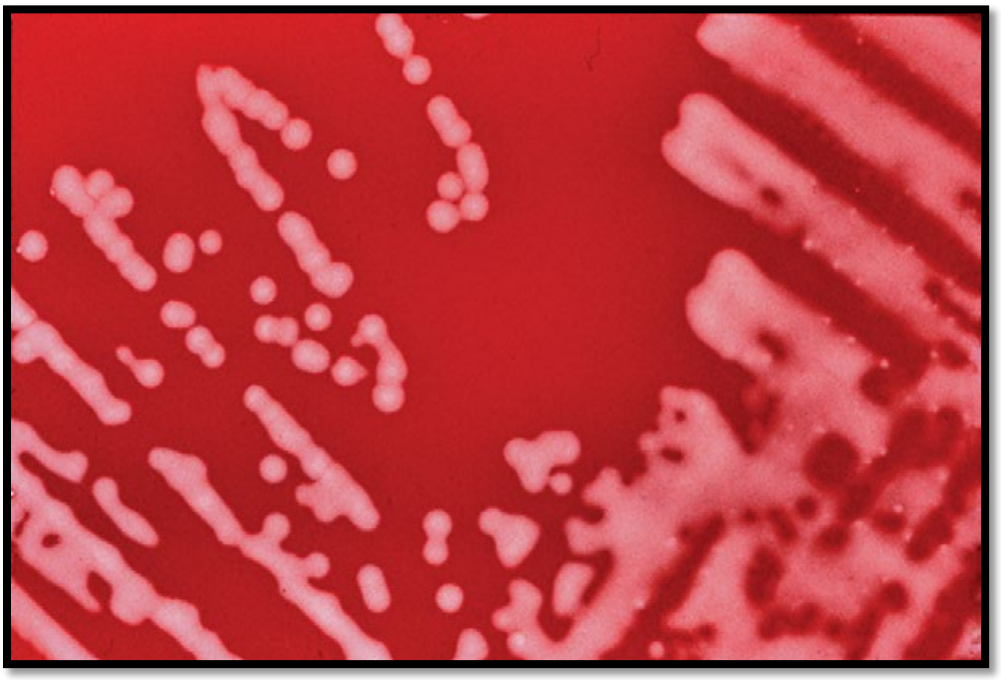


Figure 6.34 – Achromobacter xylosoxidans on SBA

- Microscopic Morphology of *Alcaligenes* spp. And *Achromobacter* spp.
    - Gram-negative bacilli possessing peritrichous flagella

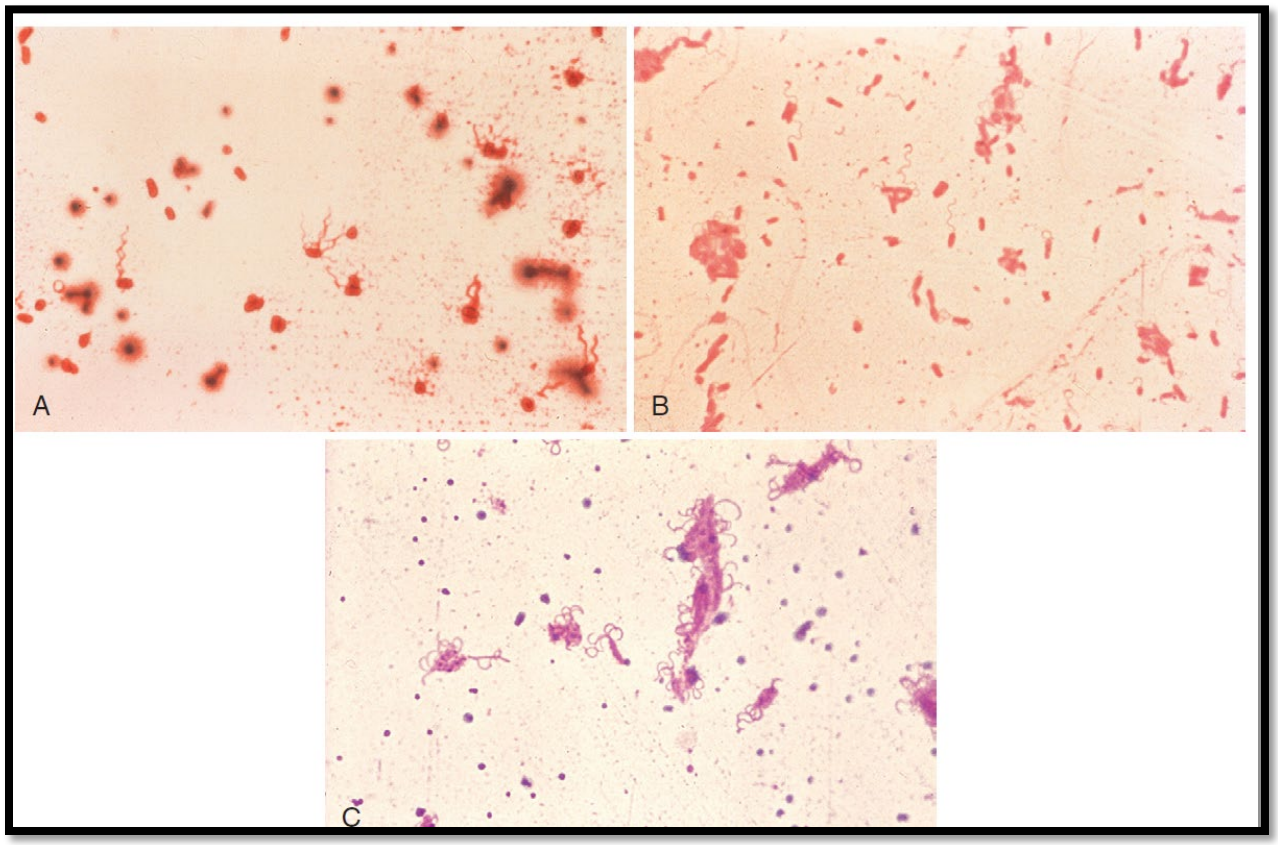


Figure 6.35 – Flagella stains of gram-negative bacilli. (A) Peritrichous flagella (e.g., A. faecalis and Achromobacter; (B) Polar, monotrichous (e.g., P. aeruginosa) (C)- Polar, multitrichous

- Biochemical Identification of *Alcaligenes* spp. *And Achromobacter* spp.
    - No oxidation in O/F media with blue color at the top
        - Except *Alcaligenes xylosoxidans*
            - Acid in glucose and xylose
    - Motile
    - Reduce nitrates to nitrites
        - *Achromobacter xylosoxidans and Alcaligenes faecalis* further reduce to nitrogen gas
    - Negative for indole
    - Negative for esculin and gelatin hydrolysis

- Antimicrobial Susceptibility *of Alcaligenes* spp. And *Achromobacter* spp.
    - *A. faecalis* and *A. piechaudii*, are usually susceptible to SXT, piperacillin, ticarcillin, ceftazidime, and quinolones
    - Resistance to aztreonam and the aminoglycosides is common
    - Some strains of *A. piechaudii* may be susceptible to amoxicillin
    - *A. xylosoxidans* is usually resistant to aminoglycosides, ampicillin, first- and second-generation cephalosporins, chloramphenicol, and fluoroquinolones
        - Usually they are susceptible to piperacillin, third generation cephalosporins, carbapenems, and SXT

Significant but rarely Encountered Organisms

- *Brevundimonas*
- *Chromobacterium violaceum*
- *Comamonas* and *Delftia*
- *Elizabethkingia meningosepticum*
- *Balneatrix alpaca*
- *Chryseobacterium indologenes*
- *Empedobacter*
- *Wautersiella*
- *Myroides odoratimimus*
- *Weeksella*
- *Sphingobacterium*
- *Methylobacterium*
- *Roseomonas mucosa*
- *Ralstonia pickettii*
- *Cupriavidus paucula*
- *Shewanella putrefaciens* and *S. algae*
- *Sphingomonas paucimobilis* and *Sphingomonas parapaucimobilis*

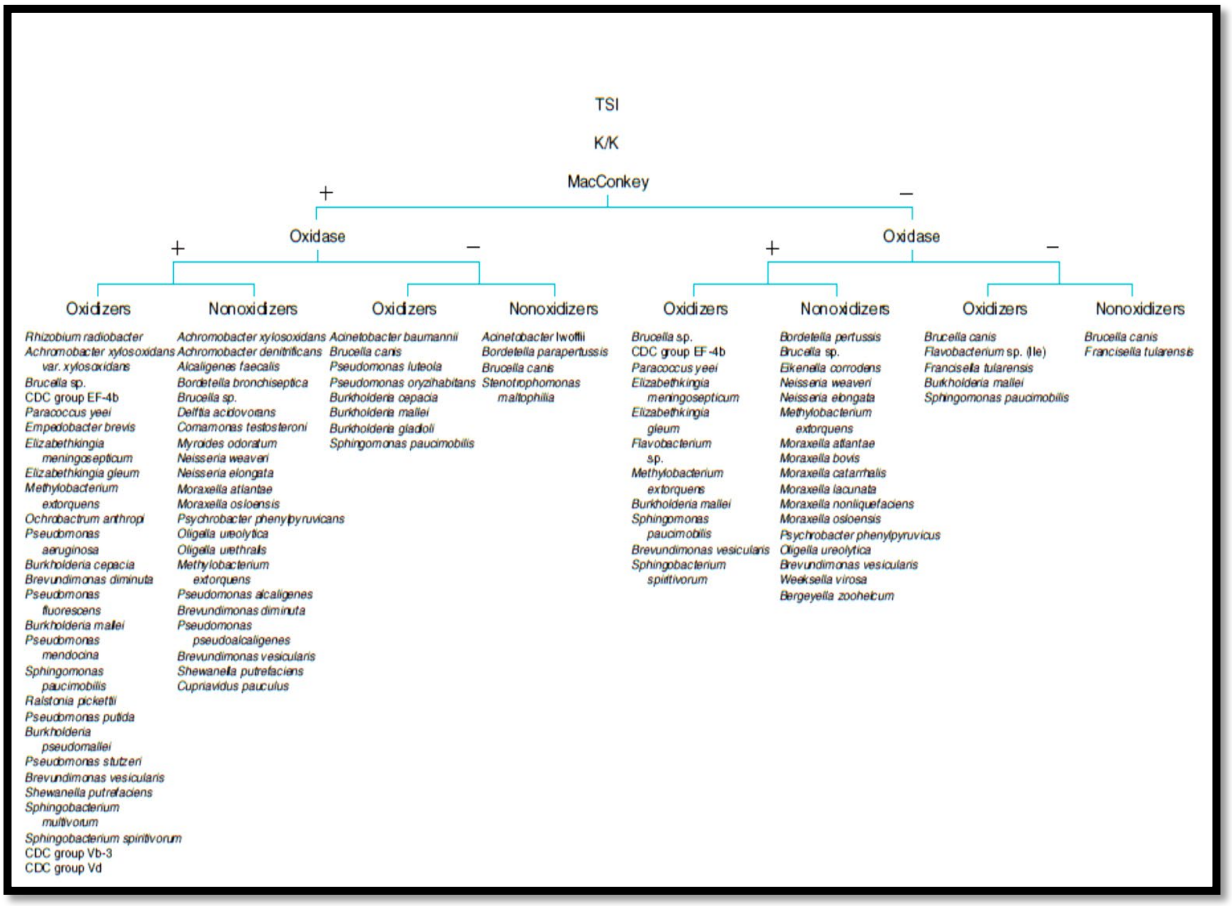
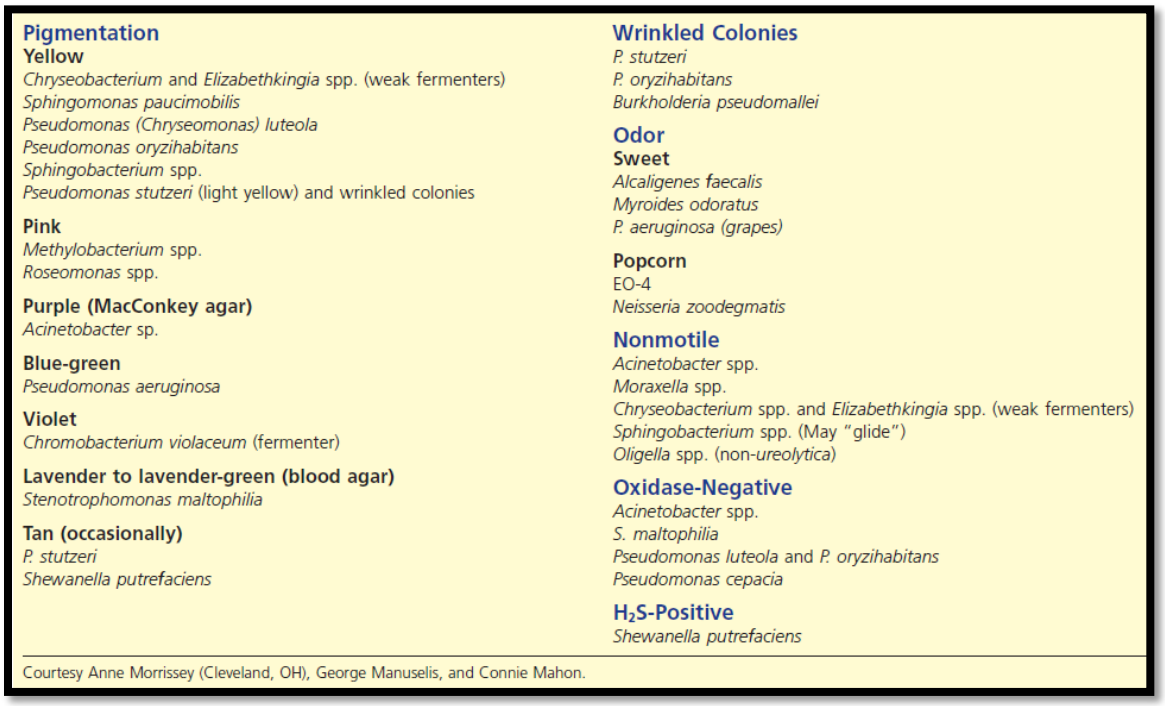


Figure 6.36 – Grouping of non-fermenters based on eight possible results. K/K, Alkaline-alkaline; TSI, triple sugar iron



Pigmentation

Yellow
Chryseobacterium and *Elizabethkingia* spp. (weak fermenters)
Sphingomonas paucimobilis
Pseudomonas (Chryseomonas) luteola
Pseudomonas oryzihabitans
Sphingobacterium spp.
Pseudomonas stutzeri (light yellow) and wrinkled colonies

Pink
Methylobacterium spp.
Roseomonas spp.

Purple (MacConkey agar)
Acinetobacter sp.

Blue-green
Pseudomonas aeruginosa

Violet
Chromobacterium violaceum (fermenter)

Lavender to lavender-green (blood agar)
Stenotrophomonas maltophilia

Tan (occasionally)
P. stutzeri
Shewanella putrefaciens

Wrinkled Colonies
P. stutzeri
P. oryzihabitans
Burkholderia pseudomallei

Odor

Sweet
Alcaligenes faecalis
Myroides odoratus
P. aeruginosa (grapes)

Popcorn
EO-4
Neisseria zoodegmatis

Nonmotile
Acinetobacter spp.
Moraxella spp.
Chryseobacterium spp. and *Elizabethkingia* spp. (weak fermenters)
Sphingobacterium spp. (May "glide")
Oligella spp. (non-*ureolytica*)

Oxidase-Negative
Acinetobacter spp.
S. maltophilia
Pseudomonas luteola and *P. oryzihabitans*
Pseudomonas cepacia

H_2S-Positive
Shewanella putrefaciens

Courtesy Anne Morrissey (Cleveland, OH), George Manuselis, and Connie Mahon.

Figure 6.37 – Characteristics Common to Groups of Non-fermenters

Checks on learning

1. What biochemical test will differentiate non-fermenters from Enterobacteriaceae (except *Plesiomonas*)?

2. What substances do non-fermenters fail to ferment in traditional media?

3. The three species of non-fermenters that make up the majority of isolates routinely seen in clinical laboratories.

Case Study #1

Patient Presentation

A 59-year-old woman was examined by her family physician for investigation for diarrhea with some mild dysentery that has occurred for about one week. The bouts include fever, gas, and often foul-smelling stools. Her reason for the visit is that, while ill, she has noticed occasional lapses of sensation in her feet as well.

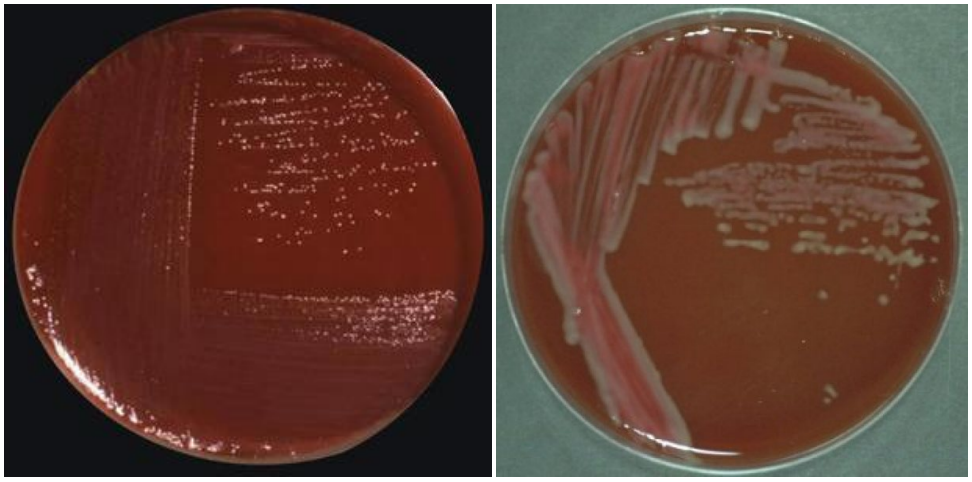
Physical Examination

Temperature of 102º F, pulse of 112/minute, blood pressure of 120/70. Tenderness of the abdomen was noted, though the feet had full sensation at the time of the visit and there was some acute chest pain as well.

Laboratory Data

Her hematocrit was 33.4, and hemoglobin was 12.2 g/100 ml. WBC count was 11.7 X 10^9. Blood cultures were drawn up as well as culturing of fecal samples. It was decided to incubate all plates at 37° and 42°C in both air and CO_2 concurrently. Growth after 48 hours was negative for the blood cultures and aerobic plates at both temperatures, though small colonies were appearing on the 42° plates that were in CO_2. It was decided that further testing was needed.

	Patient	Reference Range
WBC	11.7 x 10^9	4.5-11.5 x 10^9 /L
Hematocrit (%)	33.4	34.9-46.9
Hemoglobin (g/dl)	12.2	12.0-15.0 X 10^{12}/L



1. Based on the data presented, what organism is most likely causing the infection? What evidence supports this conclusion?

2. What biochemical tests would you perform to further identify or eliminate the organism? What are the expected results?

3. What type of susceptibility testing is performed on this organism?

4. In what type of environment does this organism grow best?

5. What disease state is associated with this organism? What is significant about it?

6. How is this organism contracted?

Case Study #2

Patient Presentation

A 21-year-old male Naval enlisted Sailor reports to a shipboard clinic on November 20 suffering from severe rice water like diarrhea and occasional spasmodic vomiting. He has only a mild fever and no other noticeable symptoms outside of weight loss and fatigue due to the diarrhea. It is noted that he was helping with supplies for victims of the super-typhoon in the Philippines and may have accidentally ingested water after falling in during unloading operations in Manila's harbor.

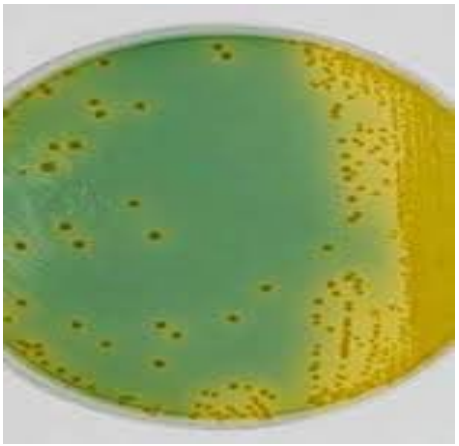
Physical Examination

The patient appeared ill, weak, and weak from fluid loss. He had visible symptoms otherwise. Vitals include temperature of 99ºf, pulse of 102/minute, blood pressure of 110/70, and 29 respirations per minute.

Laboratory Data

Urinalysis revealed normal; Hemoglobin, 15.5 g/100 ml; WBC count, 4.8 X $10^{9/L}$. Chest X-ray revealed no abnormalities. Urine, fecal, and blood samples were collected and submitted to the laboratory for bacterial culture.

	Patient	Reference Range
Hemoglobin (g/100ml)	15.5	14.0-18.0
WBC (k/ul)	4.8	4.5-11.5



1. Based on the data presented, what organism is most likely causing the infection? What evidence supports this conclusion?

2. What biochemical tests would you perform to further identify or eliminate the organism? What are the expected results?

3. What drugs are most isolates of this organism susceptible to?

4. What type of media would you use to isolate the organism? Why?

5. What disease state is associated with this organism? What is significant about it?

6. How is this organism contracted?

Case Study #3

Patient Presentation

A 62-year-old male presented with a 3-day account of suprapubic pressure, dysuria, and urgency. History included three previous episodes of urinary tract infections and prostatitis within the last 12 years and diagnosis of bilateral ureteral reflux. The third infection was treated with several different antimicrobial agents, but each failed to eliminate the infection completely.

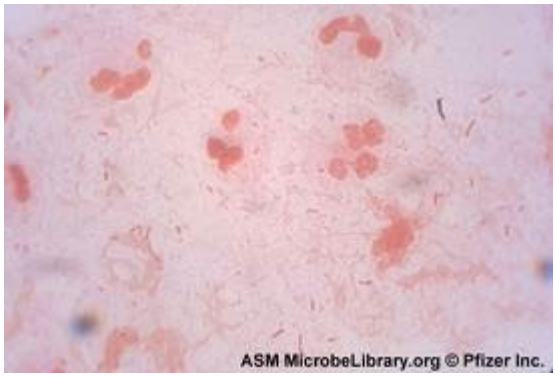
Physical Examination

Vitals include temperature of 98.8º F (37.1ºc), pulse of 65/minute, and blood pressure of 116/70. Slight enlargement of the prostate was also noted.

Laboratory Data

Urinalysis revealed 15 WBC and numerous bacteria per high power field, positive nitrate, and possible leukocyte esterase. Urine culture grew large numbers of ampicillin susceptible Gram-negative short rods on MacConkey's agar. CNA agar was negative, while BAP and Chocolate agar had abundant large, gray colonies.

	Patient		Reference Range
WBC	15		0-5





1. Based on the data presented, what organism is most likely causing the infection? What evidence supports this conclusion?

2. What biochemical tests would you perform to further identify or eliminate the organism? What are the expected results?

3. What other disease states are closely associated with this organism?

4. What type of media would you use to isolate the organism? Describe key morphological observations for each plate.

5. What disease state is the patient exhibiting? What is significant about it?

6. How is this organism contracted?

Case Study #4

Patient Presentation
A 68-year-old patient with Alzheimer disease was brought to the emergency room by the staff of a local nursing home. He presented as lethargic with a sallow complexion. During respiration, the right side of his chest moved better than the left. He showed dense consolidation of the lower lobe of the left lung on physical examination. A sputum sample revealed blood and a greenish color.

Physical Examination
Temperature of 102.4° F (39.1°c) and respiratory rate of 33/minute. Chest X-rays showed tight consolidation of the left lung with evidence of formation of cavities in the lung tissue from cytotoxic damage. The patient complained of chills in the exam room, combined with fever. A smear of his sputum demonstrated no acid-fast bacteria.

Laboratory Data
Sputum samples were plated and showed no growth on CNA, but large, translucent, spreading colonies on BAP and Chocolate agar with a sweet odor and metallic sheen. Colonies appeared on MacConkey's agar but were large and non-spreading. Gram stains showed long Gram-negative rods.



1. Based on the data presented, what organism is most likely causing the infection? What evidence supports this conclusion?

2. What biochemical tests would you perform to further identify or eliminate the organism? What are the expected results?

3. What types of drugs is this organism susceptible to? Why is this especially significant for this patient?

4. What type of media would you use to isolate the organism? Describe key morphological observations for each plate.

5. What disease state is the patient exhibiting? What is significant about it?

6. How is this organism contracted?

Case Study #5

Patient Presentation

A 4-year-old girl presents at the emergency room with bloody diarrhea, fever, and vomiting. The child's mother reports that the child has had these symptoms for about 24 hours, and she has not passed urine for about 12 hours. The child is enrolled in a day care center and the group had recently made a field trip to a fast-food place to learn about different jobs. The children had a lunch of ground beef, fries, and cola after meeting different workers. This trip was 4 days earlier on Friday.

Physical Examination

Temperature of 102.2° F (39.0°c) noted as well as physical signs of dehydration. Blood samples drawn showed evidence of greatly reduced kidney function and lysed red blood cells.

Laboratory Data

Positive blood cultures were plated to media and had no growth on CNA but pink colonies on MacConkey agar. Large gray colonies appeared on Chocolate agar. Gram stains of the colonies had short Gram-negative bacilli.



1. Based on the data presented, what organism is most likely causing the infection? What evidence supports this conclusion?

2. What cytotoxins are produced by this organism? What is significant about each one?

3. What type of assay testing is used to detect this organism?

4. What type of media would you use to isolate the organism? Describe key morphological observations for each plate.

5. What disease state is the patient exhibiting? What is significant about it?

6. How is this organism contracted?